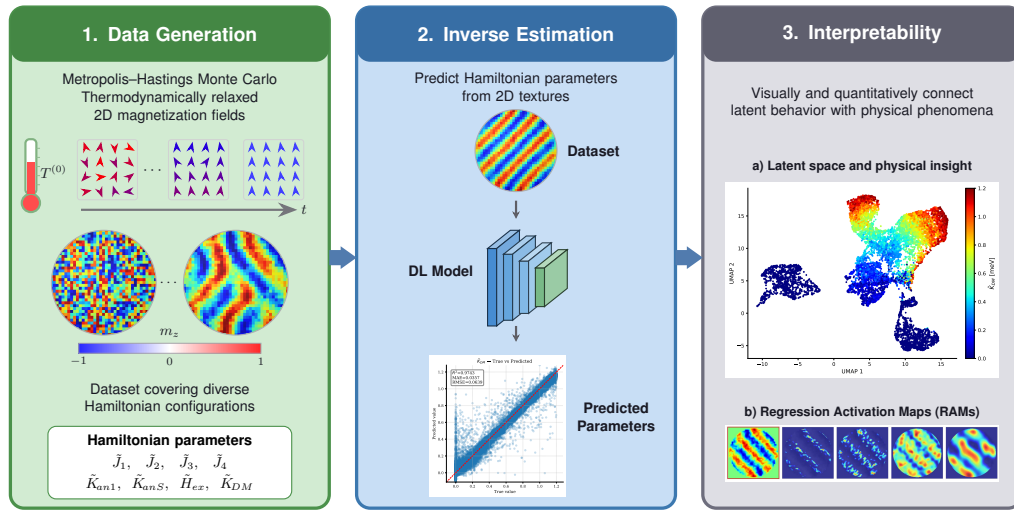


# Graphical Abstract

## Regression-Based Explainable Deep Learning for Estimating Hamiltonian Parameters from Magnetic Nanodot Images

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## Highlights

### **Regression-Based Explainable Deep Learning for Estimating Hamiltonian Parameters from Magnetic Nanodot Images**

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- Atomistic Hamiltonian generates relaxed 2D magnetic domain images for training.
- Deep learning regression recovers Hamiltonian parameters from nanodot images.
- Regression Activation Maps explain continuous prediction errors spatially.
- Framework isolates degenerate textures linking black-box AI to magnetic physics.

# Regression-Based Explainable Deep Learning for Estimating Hamiltonian Parameters from Magnetic Nanodot Images

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## Abstract

In magnetic nanodots, the stability of emergent topological textures is governed by an energetic competition described by the extended Heisenberg Hamiltonian, involving symmetric exchange, the Dzyaloshinskii–Moriya interaction (DMI), magnetocrystalline anisotropy, and the Zeeman field. Recovering these atomistic parameters from two-dimensional magnetic domain images is a severely ill-posed inverse problem, since distinct parameter combinations yield states with nearly indistinguishable free energies. Empirical fitting and simulation-based searches struggle to discriminate among competing solutions, while standard deep learning regressors collapse the multimodal inverse landscape into opaque, oversmoothed estimates. We propose an inherently explainable regression framework that couples Metropolis–Hastings Monte Carlo sampling with simulated annealing, used to generate thermodynamically relaxed magnetization fields, to convolutional and Vision Trans-

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former architectures trained to recover eight Hamiltonian parameters. Interpretability is delivered by Regression Activation Maps (RAMs), a novel attribution method extending gradient-based saliency to continuous physical targets. On 218,256 simulated configurations, the framework predicts the annealing temperature, second-shell exchange constant, and DMI strength with  $R^2$  of 0.91, 0.93, and 0.97. Unsupervised clustering of the latent space recovers four magnetic phases without supervision, and RAMs isolate chiral domain walls and skyrmion cores as the textures driving predictions, establishing a traceable link between atomistic design parameters and observable spin textures.

*Keywords:* Explainable AI, Magnetic Nanodots, Hamiltonian Parameters, Regression Activation Maps, Deep Learning

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## 1. Introduction

Understanding and controlling magnetic behavior at the nanometric scale is a central challenge in modern condensed matter physics and materials engineering [1]. This capability is particularly relevant to the development of spintronic devices, ultrahigh-density magnetic memories, and neuromorphic computing platforms based on topological magnetic textures [2, 3]. At such length scales, surface effects, geometric confinement, and energy quantization substantially modify magnetic properties relative to the bulk [4]. The resulting enhancement of surface anisotropy, structural imperfections, and thermal fluctuations strongly affects the stability and dynamics of spin configurations [5]. These effects give rise to emergent collective states—including ferromagnetic, helical, labyrinthine, and skyrmionic textures [6, 7, 8, 9]—whose stability is governed by the interplay of competing microscopic interactions [10]. In this context, magnetic nanodots constitute an ideal model platform, as their lateral confinement favors the formation of single-domain states, vortices, and skyrmion-like configurations controlled by the balance among symmetric exchange, magnetic anisotropy, DMI, and external magnetic fields [11, 12]. Therefore, inferring these underlying physical parameters directly from magnetic domain images has become a critical inverse problem for the rational engineering of functional magnetic materials [13].

Within this framework, atomistic Hamiltonian-based modeling constitutes the most rigorous description of the discrete spin interactions, localized dynamics, and boundary effects that dominate nanoscale magnetic systems.



24 Unlike continuum micromagnetic models, which can mask essential surface  
 25 and finite-size contributions, the extended Heisenberg Hamiltonian explicitly  
 26 captures the atomistic interactions responsible for stabilizing complex topo-  
 27 logical textures in magnetic nanodots [14]. In its general form, the Hamil-  
 28 tonian includes several competing terms: the symmetric exchange interac-  
 29 tion ( $\tilde{J}$  [meV]), which promotes collinear alignment; the DMI ( $\tilde{D}$  [meV]),  
 30 which favors chiral and non-collinear spin textures; the magnetocrystalline  
 31 anisotropy ( $\tilde{K}$  [meV]), which selects preferred crystallographic orientations;  
 32 and the Zeeman coupling to an external magnetic field ( $\tilde{H}$  [meV]) [15]. The  
 33 competition among these contributions generates a highly multimodal en-  
 34 ergy landscape [10, 16], in which physically distinct parameter sets may pro-  
 35 duce spin states with nearly indistinguishable free energies. This degeneracy  
 36 can lead to the coexistence of helical, skyrmion-like, and other metastable  
 37 textures [17, 18], thereby rendering the inverse estimation of Hamiltonian  
 38 parameters from two-dimensional magnetic domain images intrinsically ill-  
 39 conditioned and structurally ambiguous [19, 20].

40 The intrinsic ambiguity of the inverse problem is further exacerbated by  
 41 the limitations of current experimental and analytical methodologies. Al-  
 42 though advanced imaging techniques, including magnetic force microscopy,  
 43 Lorentz transmission electron microscopy, and spin-polarized scanning tun-  
 44 neling microscopy [21], enable high-resolution visualization of magnetic tex-  
 45 tures, they typically provide indirect or transformed measurements that are  
 46 subject to modality-dependent distortions, noise, and finite spatial resolu-  
 47 tion [22, 23]. In practice, parameter estimation therefore relies heavily on  
 48 empirical fitting or on comparison with synthetic magnetic configurations  
 49 generated by atomistic spin simulations or continuum micromagnetic mod-  
 50 els [24, 25]. Yet similarity measures defined in image space—such as mean  
 51 squared error or structural similarity—are often inadequate for resolving  
 52 topologically distinct states, since they primarily quantify visual resemblance  
 53 rather than physical equivalence. Consequently, the optimization process  
 54 remains structurally ambiguous, and substantially different Hamiltonian pa-  
 55 rameter sets may produce magnetic domain patterns that are visually indis-  
 56 tinguishable [26, 27].

57 To circumvent the prohibitive cost of iterative simulation-based optimiza-  
 58 tion, recent efforts have reformulated this inverse problem as a data-driven  
 59 learning task using deep neural architectures [28]. Although supervised re-  
 60 gression models allow fast parameter inference, they generally impose an  
 61 implicit one-to-one mapping and therefore tend to collapse the intrinsic mul-

timodality of degenerate magnetic systems into oversmoothed point predictions [29]. Methods designed to explicitly capture this uncertainty—such as Bayesian inference, diffusion models, cycle-consistent networks, and physics-informed neural networks—offer improved representational flexibility, but often introduce substantial computational overhead and pronounced optimization stiffness [30, 31, 32]. Beyond these computational challenges, a central limitation remains their opacity. In strongly degenerate parameter spaces, conventional black-box models offer little or no mechanistic understanding of how localized spatial features contribute to the inferred Hamiltonian parameters [33, 34]. Moreover, although gradient-based attribution tools such as class activation maps are widely used in classification, analogous interpretability methods for continuous regression in highly multimodal physical systems remain underdeveloped [35]. This interpretability gap restricts the physical traceability required for robust and scientifically meaningful machine-learning-based inference.

Recent advances have demonstrated the immense potential of deep learning in resolving the inverse problem of magnetic Hamiltonian parameter estimation from domain images. For example, Kwon *et al.* [20] employed ResNet architectures to estimate Dzyaloshinskii-Moriya, anisotropy, and dipolar interactions from Monte Carlo-simulated spin configurations, while Kong *et al.* [13] utilized customized convolutional neural networks to extract exchange stiffness, effective anisotropy, and interfacial DMI from micromagnetic simulations. More recently, Lee *et al.* [28] demonstrated the applicability of fully connected networks for estimating interlayer and intralayer interactions in twisted van der Waals magnets. Despite their impressive predictive accuracy, these pioneering approaches predominantly treat the neural network as a “black box.” Specifically, while some works analyze intermediate feature maps [13] or perform statistical error analysis [20, 28], they lack spatially resolved attribution mechanisms to explicitly identify which morphological features of the spin textures drive specific parameter predictions. Furthermore, physical identifiability limits—such as degenerate spin configurations arising from competing symmetric interactions—are often encountered as empirical estimation errors rather than explicitly quantified boundaries. In this work, we address these critical gaps by integrating modern deep learning architectures (DenseNet121 and Vision Transformers) with a comprehensive Explainable AI (XAI) framework. By employing Regression Activation Mapping (RAM) and UMAP/HDBSCAN clustering, we provide direct spatial mappings of the network’s attention on specific magnetic tex-

tures (e.g., skyrmions, domain walls). Moreover, we explicitly establish the physical identifiability boundaries for an 8-parameter extended Heisenberg Hamiltonian, thereby bridging the gap between high-throughput parameter regression and interpretable microscopic physics.

To address these limitations, we propose an inherently explainable framework grounded in an atomistic extended Heisenberg Hamiltonian. The proposed methodology explicitly captures the competition among symmetric exchange, DMI, magnetocrystalline anisotropy, and Zeeman coupling through Metropolis–Hastings Monte Carlo sampling and simulated annealing, enabling the generation of thermodynamically relaxed and highly accurate two-dimensional magnetization fields. These physically grounded data are subsequently integrated with a regression-based deep learning architecture for robust inverse estimation of the governing Hamiltonian parameters. The main contributions of this work are summarized as follows. First, we develop an atomistic Hamiltonian simulation framework based on the extended Heisenberg Hamiltonian, in which stochastic Monte Carlo sampling and simulated annealing are used to generate thermodynamically relaxed two-dimensional magnetization images and to establish an explicit correspondence between physical parameters and their associated topological magnetic states. Second, we integrate advanced convolutional neural networks and vision transformer architectures into a generalized regression framework for the robust prediction of fundamental Hamiltonian parameters from magnetic domain images. Third, we introduce RAMs, a novel interpretability framework that extends gradient-based activation mapping to continuous-valued physical targets, thereby providing both visual and quantitative insight into the relationship between the network’s latent representations and the underlying magnetic phenomena. Finally, we demonstrate that RAMs can isolate degenerate magnetic textures, such as chiral domain walls and skyrmions, that most strongly govern the prediction of specific Hamiltonian parameters, thus providing a physically traceable alternative to conventional deterministic models whose predictive mechanisms remain opaque.

To assess the effectiveness of the proposed framework, extensive experiments were performed using a custom-generated dataset of simulated magnetic domain images covering a broad range of Hamiltonian parameter configurations. The obtained results indicate that the proposed method enables accurate inverse prediction of key magnetic parameters, with the DenseNet121 architecture achieving the strongest overall performance among the tested models. In particular, the framework shows high predictive capability for

the DMI and symmetric exchange parameters. Moreover, the qualitative and quantitative analyses based on RAMs indicate that the learned representations are associated with physically relevant spin topologies rather than dataset-specific noise or superficial correlations, thereby strengthening the interpretability and reliability of the proposed approach. The remainder of the paper is organized as follows. Section 2 presents the related work. Section 3 details the materials and methods. Section 4 describes the experimental setup. Section 5 presents and discusses the results. Finally, Section 6 concludes the paper and highlights future research directions.

## 2. Related Work

The inverse estimation of Hamiltonian parameters from magnetic domain configurations is widely recognized as a highly challenging problem due to the complex, nonlinear relationship between physical interactions and observable spin textures. From an experimental standpoint, researchers heavily rely on advanced indirect imaging techniques such as magnetic force microscopy, Lorentz transmission electron microscopy, and spin-polarized scanning tunneling microscopy [21]. While these modalities enable high-resolution visualization of magnetic domains, they inherently capture transformed or partial projections of the underlying three-dimensional spin configurations. Because each technique is governed by distinct physical interaction principles, experimental observations are frequently contaminated by modality-specific distortions, finite spatial resolution limits, and heterogeneous noise, making direct inversion from these measurements generally infeasible [22, 23].

Consequently, practical parameter estimation heavily depends on empirical fitting or direct comparison with synthetic data generated through numerical simulations [36, 37]. Computationally, the problem is most commonly addressed using physics-based forward simulations, specifically atomistic spin models governed by the extended Heisenberg Hamiltonian and continuum micromagnetic models [24, 25, 38]. However, attempting to optimize the inverse mapping by minimizing pixel-level discrepancy functions—such as Mean Squared Error or Mean Absolute Error—frequently fails to capture physically meaningful topological differences or vital spin interactions [39, 40]. Even structural metrics like the Structural Similarity Index fall short in this regard [27]. Ultimately, because the comparison between simulated and experimental data occurs strictly in the image space rather than the parameter space, the optimization landscape becomes highly ambigu-

ous, allowing disparate parameter configurations to yield visually identical domain patterns [26].

To bypass the computationally prohibitive nature of iterative simulation-based optimization, recent advances have increasingly framed the inverse problem as a data-driven mapping utilizing deep learning architectures [28]. The most prevalent approach employs supervised regression models, including convolutional neural networks and transformer-based architectures, to approximate this mapping deterministically. While these models offer rapid inference and scalability, they inherently assume a one-to-one mapping between images and parameters. In highly degenerate magnetic systems, this assumption causes the models to collapse the intrinsic multimodality of the parameter space into a single, often smoothed, point estimate, rendering them highly sensitive to dataset bias and sim-to-real mismatch [29].

To explicitly model uncertainty and capture this multimodality, probabilistic frameworks such as Bayesian inference methods, conditional variational autoencoders, and diffusion models have been introduced [30, 41, 42, 43]. Despite their theoretical flexibility in sampling multiple plausible configurations, these generative and Bayesian approaches suffer from immense computational overhead, unstable training dynamics, and a distinct lack of explicit physical guarantees. In parallel, cycle-consistent frameworks and Physics-Informed Neural Networks attempt to constrain the solution space by enforcing coherence between forward and inverse mappings, or by directly embedding thermodynamic penalties into the loss function [31, 44, 32]. While physically plausible, these hybrid models introduce severe optimization stiffness and rely fundamentally on the accuracy of the underlying forward surrogate, which can dangerously propagate errors within strongly degenerate spin regimes.

Despite the growing predictive capabilities of advanced deep neural networks, their fundamental opacity remains a critical barrier to their widespread adoption in condensed matter physics. In the context of nanoscale magnetism, the delicate energetic competition among the exchange interaction, the DMI, and magnetocrystalline anisotropy creates a densely degenerate parameter space where multiple physical configurations present nearly identical free energies [34, 45]. Traditional black-box deep learning architectures provide no mechanistic insight into how specific spatial features within a magnetic domain image correlate with the predicted Hamiltonian constants [33, 46]. This lack of interpretability prevents researchers from distinguishing whether a model has genuinely captured the underlying physical

laws or merely memorized superficial dataset artifacts. While gradient-based attribution methods like Class Activation Maps have been widely adopted for discrete classification tasks, interpreting continuous regression outputs in highly multimodal physical systems remains largely underexplored [35]. The inability to spatially trace and quantify the sources of prediction errors restricts the scientific traceability of these models, underscoring an urgent necessity for architectures that inherently couple high predictive accuracy with transparent, physics-informed spatial interpretations.

To overcome the limitations, our methodology is fundamentally grounded in a rigorous atomistic extended Heisenberg Hamiltonian framework. By explicitly modeling the delicate energetic competition among symmetric exchange, DMI, magnetocrystalline anisotropy, and Zeeman coupling through Metropolis-Hastings Monte Carlo sampling and simulated annealing, we computationally stabilize and extract highly accurate, thermodynamically relaxed two-dimensional spatial magnetization fields. In contrast to prevalent black-box methodologies and computationally prohibitive probabilistic frameworks, we directly couple these physically exhaustive simulations with an inherently explainable, regression-based deep learning architecture explicitly designed for the rigorous inverse estimation of these governing parameters. By integrating advanced Convolutional and Visual Transformer networks, we robustly tackle the structural ambiguity of the inverse problem without sacrificing architectural transparency. Also, we introduce a RAMs, an interpretability approach that directly correlates localized spatial activations within the simulated magnetic domain visualizations to the absolute prediction error of the continuous physical variables. Unlike standard structural metrics or deterministic regression models that obscure the origin of their predictions, our method visually and quantitatively bridges the latent behavior of the neural network with the rigorously modeled phenomena. This enables the precise isolation of degenerate magnetic textures—such as chiral domain walls and skyrmions—that drive specific parameter estimations, establishing a highly reliable, transparent, and traceable pathway for analyzing complex nanoscale magnetic systems.

### 3. Materials and Methods

#### 3.1. Atomistic Hamiltonian Framework

The theoretical foundation for modeling interacting magnetic dipole moments originates from the seminal formulations of Ernst Ising and Werner

248 Heisenberg [47]. Ising introduced a one-dimensional statistical mechanics  
 249 model wherein localized spins were heavily constrained to discrete, scalar  
 250 states (i.e., rigidly pointing “up” or “down”). While the Ising model pro-  
 251 vided early fundamental insights into macroscopic ferromagnetism and phase  
 252 transitions, its strict uni-axial constraint renders it structurally insufficient  
 253 for capturing the continuous rotational degrees of freedom required to model  
 254 complex, multi-dimensional magnetic textures. This limitation was subse-  
 255 quently resolved by Heisenberg, who formulated a generalized model allowing  
 256 spin vectors to vary continuously in three-dimensional space with rotational  
 257 symmetry. While the foundational Heisenberg model successfully captures  
 258 isotropic exchange driven by the Pauli exclusion principle, modern nanoscale  
 259 applications—such as the study of magnetic vortices and topologically pro-  
 260 tected skyrmions in confined nanodot geometries—necessitate an extended  
 261 approach [3]. This extended framework rigorously incorporates symmetry-  
 262 breaking relativistic spin-orbit coupling effects, primarily the DMI and mag-  
 263 netocrystalline anisotropy, alongside standard Zeeman coupling [48].

264 Formally, let  $\mathcal{V} = \{1, \dots, N\}$  denote the set of indices for the discrete  
 265 magnetic lattice sites. The thermodynamic state of the system is character-  
 266 ized by the spatial distribution of localized magnetic moments, represented  
 267 by the normalized spin vectors  $\mathbf{s}_i \in \mathbb{R}^3$ , subject to the constraint  $\|\mathbf{s}_i\|_2 = 1$ ,  
 268 for  $i \in \mathcal{V}$ . In a generalized theoretical context, all pairwise symmetric and  
 269 antisymmetric interactions, as well as site-specific field and anisotropy align-  
 270 ments, can be collapsed into a unified geometric approach [49, 50]. The total  
 271 effective energy of the system  $\mathcal{H} \in \mathbb{R}$  [meV] is thus defined by the following  
 272 generalized quadratic form:

$$\mathcal{H}(\mathbf{S}|\tilde{\theta}) = - \sum_{i=1}^N \sum_{j \in \mathcal{N}(i)} \langle \mathbf{s}_i, \tilde{\mathbf{M}}_{ij} \mathbf{s}_j \rangle - \sum_{i=1}^N \langle \tilde{\mathbf{h}}_i, \mathbf{s}_i \rangle - \sum_{i=1}^N \langle \mathbf{s}_i \tilde{\mathbf{K}}_i, \mathbf{s}_i \rangle, \quad (1)$$

273 where  $\mathcal{N}(i)$  defines the set of interacting nearest neighbors for a given site  
 274  $i$ , and the matrix  $\mathbf{S} \in \mathbb{R}^{N \times 3}$  gathers the continuous spin vectors across the  
 275 lattice with  $\mathbf{s}_i \in \mathbf{S}$ . Here,  $\tilde{\theta} = \{\tilde{\mathbf{M}}_{ij}, \tilde{\mathbf{h}}_i, \tilde{\mathbf{K}}_i\}$ , where  $\tilde{\mathbf{M}}_{ij} \in \mathbb{R}^{3 \times 3}$  represents  
 276 the complete bilinear interaction matrix coupling neighboring spins,  $\tilde{\mathbf{h}}_i \in \mathbb{R}^3$   
 277 accounts for linear external field interactions, and  $\tilde{\mathbf{K}}_i \in \mathbb{R}^{3 \times 3}$  is the anisotropy  
 278 tensor governing site-specific directional preferences.

279 Afterward, to recover the isolated physical phenomena and define an ex-  
 280 tended target scalar parameter space  $\boldsymbol{\theta} = [\tilde{J}_1, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{D}, \tilde{H}, \tilde{K}_{an1}, \tilde{K}_{anS}]$ ,

the generalized matrices must be mathematically decomposed. By factorizing these parameters, we can map each matrix operation to its specific energetic contribution:

- *Bilinear Coupling* ( $\tilde{\mathbf{M}}_{ij}$ ): A real square matrix can be separated into its symmetric and skew-symmetric components. Here,  $\tilde{\mathbf{M}}_{ij}$  is parameterized by factorizing the isotropic exchange scalar  $\tilde{J}_r$  for the  $r$ -th neighbor shell and the DMI magnitude  $\tilde{D}$  along a predefined spatial direction vector  $\hat{\mathbf{d}}_{ij} \in \mathbb{R}^3$  ( $\|\hat{\mathbf{d}}_{ij}\|_2 = 1$ ), yielding:

$$\tilde{\mathbf{M}}_{ij}(\tilde{J}_r, \tilde{D}) = \tilde{J}_r \mathbf{I} - \tilde{D}[\hat{\mathbf{d}}_{ij}]_{\times}, \quad (2)$$

where  $\mathbf{I} \in \mathbb{R}^{3 \times 3}$  represents the identity matrix and  $[\hat{\mathbf{d}}_{ij}]_{\times} \in \mathbb{R}^{3 \times 3}$  denotes the skew-symmetric matrix representation of the cross-product. The symmetric exchange is modeled across four neighbor shells  $r \in \{1, 2, 3, 4\}$ , where  $\mathcal{N}_r(i)$  denotes the set of neighbors at shell  $r$  around site  $i$ , ordered by increasing interatomic distance ( $d_1 < d_2 < d_3 < d_4$ ). To resolve the fundamental scale ambiguity of the inverse problem—whereby multiplying all parameters by a common factor leaves the observable spin configuration unchanged—the first-shell exchange is fixed as the energy reference unit,  $\tilde{J}_1 \in \mathbb{R}$  in meV, while  $\tilde{J}_2, \tilde{J}_3, \tilde{J}_4 \in \mathbb{R}$  are free parameters (also in meV). The DMI acts exclusively between first-shell neighbors and is therefore indexed only over  $\mathcal{N}_1(i)$ .

- *Zeeman Interaction* ( $\tilde{\mathbf{h}}_i$ ): The linear field term is factorized by extracting the scalar field magnitude  $\tilde{H}$  and the fixed normalized direction vector  $\hat{\mathbf{h}} = \hat{\mathbf{e}}_x$ , with  $\|\hat{\mathbf{e}}_x\|_2 = 1$ , of the externally applied magnetic field, yielding  $\tilde{\mathbf{h}}_i = \tilde{H} \hat{\mathbf{e}}_x \in \mathbb{R}^3$ . Fixing the field direction along  $\hat{\mathbf{e}}_x$  eliminates the orientational degree of freedom of  $\tilde{\mathbf{h}}_i$ , ensuring that  $\tilde{H}$  alone serves as the free parameter governing the Zeeman contribution.
- *Magnetocrystalline Anisotropy* ( $\tilde{\mathbf{K}}_i$ ): Modeling cubic magnetocrystalline anisotropy requires factorizing a site-dependent scalar constant  $\tilde{K}_i^{eff}$  to scale a quartic interaction. The principal crystallographic axes of the nanodot are defined by the orthonormal basis vectors  $\mathbf{e}_{\hat{x}}, \mathbf{e}_{\hat{y}}, \mathbf{e}_{\hat{z}} \in \mathbf{E}$ , with  $\mathbf{E} \in \mathbb{R}^{3 \times 3}$ , and the directional cosines are given by  $s_{im} = \langle \mathbf{e}_m, \mathbf{s}_i \rangle$  ( $\forall m \in \{\hat{x}, \hat{y}, \hat{z}\}$ ). Because atoms at the nanodot surface experience broken inversion symmetry—having fewer neighbors on the exterior side—their anisotropy differs fundamentally from that of interior atoms. The



lattice is therefore partitioned into two disjoint subsets: the bulk region  $\mathcal{V}_{bulk} \subset \mathcal{V}$ , comprising interior sites with complete coordination shells, and the surface region  $\mathcal{V}_{surf} \subset \mathcal{V}$ , comprising boundary sites exposed to the geometric confinement, with  $\mathcal{V}_{bulk} \cup \mathcal{V}_{surf} = \mathcal{V}$  and  $\mathcal{V}_{bulk} \cap \mathcal{V}_{surf} = \emptyset$ . This yields the site-dependent effective anisotropy constant:

$$\tilde{K}_i^{eff} = \begin{cases} \tilde{K}_{an1} & \text{if } i \in \mathcal{V}_{bulk}, \\ \tilde{K}_{anS} & \text{if } i \in \mathcal{V}_{surf}, \end{cases} \quad (3)$$

where  $\tilde{K}_{an1}, \tilde{K}_{anS} \in \mathbb{R}$  are scalar constants measured in meV per atom [51].

Substituting these factorized relationships back into Equation 1 yields the following explicit formulation for the total extended Hamiltonian:

$$\begin{aligned} \mathcal{H}(\mathbf{S}|\hat{\mathbf{d}}_{ij}, \hat{\mathbf{e}}_x; \boldsymbol{\theta}) = & - \sum_{r=1}^4 \tilde{J}_r \sum_{i=1}^N \sum_{j \in \mathcal{N}_r(i)} \langle \mathbf{s}_i, \mathbf{s}_j \rangle + \tilde{D} \sum_{i=1}^N \sum_{j \in \mathcal{N}_1(i)} \langle \hat{\mathbf{d}}_{ij}, (\mathbf{s}_i \times \mathbf{s}_j) \rangle \\ & - \tilde{H} \sum_{i=1}^N \langle \hat{\mathbf{e}}_x, \mathbf{s}_i \rangle + \frac{1}{S_o^4} \sum_{i=1}^N \tilde{K}_i^{eff} (s_{ix}^2 s_{iy}^2 + s_{ix}^2 s_{iz}^2 + s_{iy}^2 s_{iz}^2), \end{aligned}$$

with  $\tilde{J}_1 = 1.0$  meV fixed as the energy reference unit. The cubic anisotropy form in the last term follows directly from expanding  $\frac{1}{2}(1 - \|\mathbf{s}_i \mathbf{E}\|_4^4)$  under the  $L_2$  normalization constraint  $\|\mathbf{s}_i\|_2 = 1$ , which yields  $s_{ix}^2 s_{iy}^2 + s_{iy}^2 s_{iz}^2 + s_{iz}^2 s_{ix}^2$ , normalized by  $S_o^4$  to maintain dimensional consistency across spin magnitudes.

Physically, the extended parameter space  $\boldsymbol{\theta}$  actively governs the competing energetic contributions that shape the magnetic energy landscape. Each  $\tilde{J}_r$  dictates the collinear alignment tendency between spin pairs separated by the corresponding distance [52]: positive values ( $\tilde{J}_r > 0$ ) favor parallel, ferromagnetic alignment, while negative values ( $\tilde{J}_r < 0$ ) drive antiparallel, antiferromagnetic order. This multi-shell formulation is essential in confined nanodot geometries, where longer-range exchange pathways—mediated through intermediate atomic orbitals—contribute non-negligibly to the global energetic competition and can selectively stabilize or destabilize complex topological textures beyond what first-shell interactions alone would predict [50]. Among these constants,  $\tilde{J}_1$  plays a privileged role: it corresponds to the strongest and most spatially direct exchange pathway, coupling each

spin to its nearest neighbors in the cubic lattice and establishing the dominant energy scale of the magnetic system. For this reason,  $\tilde{J}_1$  is fixed to 1.0 meV throughout all simulations, serving as the absolute energy reference unit against which all remaining interaction strengths are expressed. The remaining constants  $\tilde{J}_2$ ,  $\tilde{J}_3$ , and  $\tilde{J}_4$  are free parameters whose magnitudes decay with increasing shell distance, consistent with the exponential attenuation of quantum mechanical exchange integrals with interatomic separation [52].

The DMI magnitude, represented by  $\tilde{D}$ , is responsible for driving chiral, non-collinear spin winding in systems exhibiting broken spatial inversion symmetry [53]. By actively competing with the dominant first-shell exchange  $\tilde{J}_1$ , this antisymmetric contribution stabilizes complex topological textures, such as chiral domain walls and skyrmions. It acts exclusively between first-shell neighbors, as longer-range antisymmetric exchange pathways are negligibly small compared to the dominant spin-orbit coupling at nearest-neighbor interfaces. Measured in meV per interacting spin pair, its magnitude is fundamentally dependent on the strength of the spin-orbit coupling at the material's interfaces, with typical values spanning  $[-0.5, 0.5]$  meV as a fraction of the first-shell exchange energy [53].

The Zeeman coupling energy, parameterized as  $\tilde{H}$ , quantifies the interaction between the localized atomic spins and the externally applied magnetic field [14]. It systematically penalizes magnetic misalignment with the fixed crystallographic axis  $\hat{\mathbf{e}}_x$ , frequently serving as the primary thermodynamic driving force for macroscopic magnetic phase transitions. This parameter is expressed in meV per atom, mathematically representing the effective energy equivalence ( $g\mu_B B$ ) that transitions a macroscopic magnetic field  $B$  measured in Tesla into the atomistic energy regime. Its boundaries range from 0.0 meV to approximately 2.0 meV, directly correlating to laboratory-scale magnetic fields between 0 and 10 Tesla. Also, fixing the field direction along  $\hat{\mathbf{e}}_x$  reduces the dimensionality of the inverse problem by eliminating the orientational degree of freedom of  $\tilde{\mathbf{h}}_i$ , ensuring that  $\tilde{H}$  alone governs the Zeeman contribution as a single identifiable scalar.

Finally, the magnetocrystalline anisotropy encodes the intrinsic directional preference for magnetization relative to the underlying crystal lattice [51]. Arising from relativistic spin-orbit coupling, this term rigorously establishes the energetically favorable (easy) and unfavorable (hard) crystallographic axes for spin orientation within the nanodot. Both constants are measured in meV per atom and represent a relatively subtle symmetry-breaking energy compared to the dominant isotropic exchange. The bulk

377 anisotropy constant  $\tilde{K}_{an1}$  governs interior lattice sites  $i \in \mathcal{V}_{bulk}$ , where each  
 378 atom is fully coordinated by neighbors in all spatial directions, yielding a  
 379 locally symmetric energetic environment. Its conventional boundaries range  
 380 from roughly 0.001 meV in soft bulk magnets to 1.0 meV in highly anisotropic  
 381 hard nanomaterials [51]. The surface anisotropy constant  $\tilde{K}_{anS}$ , by contrast,  
 382 governs boundary sites  $i \in \mathcal{V}_{surf}$ , where broken spatial inversion symmetry—  
 383 arising from the absence of neighbors on the exterior side of the nanodot—  
 384 produces an asymmetric local environment that intensifies the directional  
 385 preference for spin orientation. Consequently,  $|\tilde{K}_{anS}|$  is generally larger than  
 386  $|\tilde{K}_{an1}|$ , reflecting the enhanced symmetry-breaking at the nanodot bound-  
 387 ary. This two-constant formulation is essential for accurately capturing the  
 388 finite-size and surface-dominated effects that critically govern the stability  
 389 of emergent topological textures in confined nanoscale geometries [3, 48].

### 390 3.2. Image-Based Dataset Generation From Monte Carlo Simulations

391 To accurately capture the physical thermodynamic behavior and algo-  
 392 rithmically [drive the system toward a low-energy, thermodynamically stable](#)  
 393 [configuration](#) of the nanodot, thermal fluctuations must be formally intro-  
 394 duced through the framework of statistical mechanics. While the atomistic  
 395 Hamiltonian  $\mathcal{H}$  rigorously defines the internal magnetic energy for a given  
 396 spatial configuration of spins, it evaluates the system strictly as a frozen, zero-  
 397 temperature snapshot. Hence, temperature is fundamentally a macroscopic  
 398 parameter that governs the statistical distribution of the discrete magnetic  
 399 microstates rather than acting as a localized interaction within the Hamilto-  
 400 nian. Consequently, temperature  $T$  [K] couples to the system via the Boltz-  
 401 mann distribution [50]. Formally, a magnetic microstate  $\mu$  represents a spe-  
 402 cific, instantaneous realization of the continuous spin matrix, denoted as  $\mathbf{S}_\mu$ .  
 403 The total thermodynamic energy of this specific configuration is evaluated  
 404 against the static environmental vectors and physical interaction strengths,  
 405 such that  $\mathcal{H}(\mu) \equiv \mathcal{H}(\mathbf{S}_\mu | \hat{\mathbf{d}}_{ij}, \hat{\mathbf{h}}, \mathbf{E}; \boldsymbol{\theta})$ . At thermal equilibrium, the probab-  
 406 ility  $P(\mu)$  of the system occupying this specific microstate is defined as:

$$P(\mu) = \frac{1}{\mathcal{Z}} \exp\left(-\frac{\mathcal{H}(\mu)}{k_B T}\right), \quad (4)$$

407 where  $k_B$  is the Boltzmann constant (expressed in meV/K). The term  $\mathcal{Z}$   
 408 represents the canonical partition function, which mathematically enforces  
 409 probability normalization by summing over all valid spin configurations  $\nu$

410 (i.e., all possible state matrices  $\mathbf{S}_\nu$  strictly satisfying the geometric boundary  
 411 constraint  $\|\mathbf{s}_i\|_2 = 1$ ):

$$\mathcal{Z} = \sum_{\mathbf{S}_\nu} \exp \left( -\frac{\mathcal{H}(\mathbf{S}_\nu | \hat{\mathbf{d}}_{ij}, \hat{\mathbf{h}}, \mathbf{E}; \boldsymbol{\theta})}{k_B T} \right). \quad (5)$$

412 In our computational framework, this thermodynamic distribution is ac-  
 413 tively sampled using the Metropolis-Hastings Monte Carlo algorithm com-  
 414 bined with a simulated annealing protocol [54]. This stochastic sampling  
 415 ensures a rigorous exploration of the highly multimodal energy landscape  
 416 characteristic of these degenerate magnetic systems. When a localized spin  
 417 perturbation proposes a physical transition from the current microstate  $\mu^{(t)}$   
 418 (characterized by  $\mathbf{S}_{\mu^{(t)}}$ ) to a newly modified state  $\mu^{(t+1)}$  (characterized by  
 419  $\mathbf{S}_{\mu^{(t+1)}}$ ), the resulting change in the system's total energy is computed as the  
 420 scalar difference  $\Delta\mathcal{H} = \mathcal{H}(\mu^{(t+1)}) - \mathcal{H}(\mu^{(t)})$ . To optimize computational effi-  
 421 ciency,  $\Delta\mathcal{H}$  is evaluated locally at the perturbed lattice site. The proposed  
 422 state is subsequently accepted or rejected based on the precise transition  
 423 probability:

$$P(\mu^{(t)} \rightarrow \mu^{(t+1)}) = \begin{cases} 1 & \text{if } \Delta\mathcal{H} < 0 \\ \exp\left(-\frac{\Delta\mathcal{H}}{k_B T^{(t)}}\right) & \text{if } \Delta\mathcal{H} \geq 0. \end{cases} \quad (6)$$

424 Physically, the instantaneous temperature parameter  $T^{(t)}$  actively scales  
 425 the energetic barriers within the system's vast topological energy landscape  
 426 during the annealing process [54]. At elevated temperatures, the available  
 427 thermal energy  $k_B T^{(t)}$  frequently exceeds local energy penalties ( $\Delta\mathcal{H}$ ), allow-  
 428 ing the simulation to accept thermodynamically unfavorable spin alignments.  
 429 This thermal noise provides the critical kinetic mechanism required to break  
 430 out of metastable, localized energy minima, such as misaligned ferromagnetic  
 431 domains [55]. As  $T^{(t)}$  is systematically scaled down during the annealing pro-  
 432 cedure, the probability of accepting higher-energy configurations undergoes  
 433 exponential decay. This controlled thermal relaxation drives the continuous  
 434 spin vector field toward a low-energy, thermodynamically stable configuration  
 435 consistent with the annealing schedule, promoting the stabilization of com-  
 436 plex chiral skyrmion textures driven by the underlying DMI. We note that  
 437 while simulated annealing with a finite cooling rate does not formally guar-  
 438 antee convergence to the absolute global minimum—particularly in DMI-  
 439 driven systems where numerous metastable chiral states may coexist—the

monotonic energy decrease observed across all simulated configurations (see Section 5) provides empirical evidence of thorough thermodynamic relaxation within the sampled parameter space.

Moreover, to ensure that each simulation consistently originates from a fully randomized paramagnetic phase regardless of the specific parameter configuration, the initial annealing temperature (i.e.,  $T^{(t)}$  at  $t = 0$ ) is established as an *exchange-normalized initial temperature*. Consistent with foundational statistical mechanics [50], this starting temperature is dynamically rescaled in direct proportion to the dominant symmetric exchange energy ( $\tilde{J}$ ). This energetic normalization guarantees that the initial thermal fluctuations are strictly sufficient to overcome the highest localized energetic barriers ( $\Delta\mathcal{H}$ ) for that specific Hamiltonian. This effectively erases initial state bias and prevents the continuous spin vector field from prematurely collapsing into metastable local minima before the controlled cooling protocol begins.

To generate a robust and diverse dataset for deep learning evaluation, the magnetic parameters governing the Hamiltonian were systematically varied across the simulations. Key variables included the exchange interaction constant ( $\tilde{J}$ ), the magnetocrystalline anisotropy ( $\tilde{K}$ ), the external Zeeman field ( $\tilde{H}$ ), and the DMI strength ( $\tilde{D}$ ), alongside variations in the base initial temperature ( $T^{(0)}$ ). Subsequently, the transformation of these raw, converged 3D spin states into normalized 2D image arrays followed a strict procedural pipeline:

- *Geometry parsing:* Spatial coordinates defining the nanodot geometry (with a fixed disk radius and thickness as Magnetic Unit Cell (MUC) measure) were established using a binary circular lattice mask.
- *Spin state mapping:* The topological structure was projected into two dimensions by extracting the out-of-plane  $s_z$  component of the continuous spin vectors ( $\mathbf{s}_i$ ) across the valid lattice coordinates.
- *Image rendering:* The isolated configurations were rasterized onto a fixed  $\tilde{H} \times \tilde{W}$  pixel grid. A divergent red–blue colormap was utilized to strictly encode opposite spin polarities, visually capturing the chiral winding from the nanodot’s core to its perimeter.
- *Preprocessing:* Pixel intensities corresponding to the  $S_z$  magnitude were normalized to a strict interval  $[0, 1]$  to ensure stable gradient flows

475 during the deep learning optimization phase. To preserve the intrinsic  
 476 physical symmetries and chiral handedness dictated by the DMI, no  
 477 spatial augmentations (such as random rotations or structural flips)  
 478 were applied to the dataset.

479 Lastly, the simulation pipeline yields a comprehensive image-based dataset,  
 480 formalized as  $\mathcal{D} = \{\mathbf{X}_n, \boldsymbol{\theta}_n\}_{n=1}^{\check{N}}$ , which serves as the final collection of input-  
 481 output pairs for the deep learning parameter estimation task. Each input ten-  
 482 sor  $\mathbf{X}_n \in [0, 1]^{\check{H} \times \check{W}}$  encodes the two-dimensional spatial magnetization field  
 483 (the normalized  $s_z$  projection), while the corresponding target vector  $\boldsymbol{\theta}_n \in \mathbb{R}^{\check{P}}$   
 484 stores the  $\check{P}$ -dimensional array of generating thermodynamic and Hamilto-  
 485 nian parameters (e.g.,  $\boldsymbol{\theta}_n = [T_n^{(0)}, \tilde{J}_{1n}, \tilde{J}_{2n}, \tilde{J}_{3n}, \tilde{J}_{4n}, \tilde{D}_n, \tilde{H}_n, \tilde{K}_{an1n}, \tilde{K}_{anSn}]$ ).  
 486 Figure 1 depicts the main flowchart of our image-based magnetic dataset  
 487 generation approach.

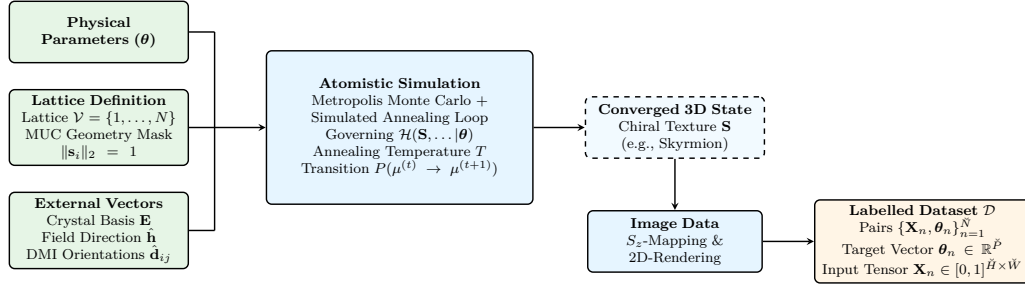


Figure 1: Main flowchart detailing the generation of the image-based magnetic dataset. The pipeline initializes with the foundational inputs (green), encompassing the Hamiltonian interaction parameters ( $\boldsymbol{\theta}$ ), the discrete unit cell geometry, and the external directional vectors. These constraints govern the atomistic simulation (blue), which employs the Metropolis Monte Carlo algorithm coupled with simulated annealing to thermodynamically relax the system into a converged 3D topological state. Finally, the resulting continuous vector field is projected, rendered, and preprocessed (yellow) to formulate the labelled dataset  $\mathcal{D}$ , mapping 2D spatial magnetization images to their generating physical parameters.

### 488 3.3. Deep Regression Framework for Hamiltonian Parameter Estimation

489 The inverse estimation of the Hamiltonian parameters and thermody-  
 490 namic state from magnetic domain images is formalized as a continuous  
 491 mapping problem. Given the generated dataset  $\mathcal{D}$ , where  $\mathbf{X}_n$  represents  
 492 the normalized  $s_z$  spatial magnetization field and  $\boldsymbol{\theta}_n$  represents the vector

of generating physical parameters, we aim to learn a parameterized non-linear mapping function  $f_{\Theta} : \mathbb{R}^{\check{H} \times \check{W}} \rightarrow \mathbb{R}^{\check{P}}$ , to find the predicted parameters  $\hat{\boldsymbol{\theta}}_n \in \mathbb{R}^{\check{P}}$ .

Here,  $f_{\Theta}$  denotes a deep neural network architecture parameterized by a set of learnable weights and biases  $\Theta$ . The optimal parameter set  $\hat{\Theta}$  is obtained by minimizing a continuous loss function over the training distribution [56]:

$$\hat{\Theta} = \arg \min_{\Theta} \frac{1}{\check{N}} \sum_{n=1}^{\check{N}} \mathcal{L}(\boldsymbol{\theta}_n, f_{\Theta}(\mathbf{X}_n)), \quad (7)$$

where  $\mathcal{L}$  represents the chosen regression objective, typically the Mean Squared Error (MSE) or Huber loss, which mitigates the influence of outlier topological states during optimization. By minimizing this objective, the network intrinsically learns to untangle the highly multimodal spatial features mapping to the specific energetic balances of the extended Heisenberg Hamiltonian.

To achieve robust feature extraction across varying scales of magnetic textures (e.g., from localized domain walls to global skyrmion geometries), this framework evaluates four distinct, state-of-the-art deep learning architectures: ResNet50, DenseNet, Xception, and the Visual Transformer (ViT).

– *Residual Networks (ResNet50)* [57]: Standard deep convolutional networks often suffer from the vanishing gradient problem as depth increases, hindering the learning of complex physical mappings. ResNet50 circumvents this by introducing shortcut connections that bypass multiple convolutional layers, directly propagating spatial information and gradients. Mathematically, if  $\mathbf{x}$  is the input to a sequence of layers, and  $\mathcal{F}(\mathbf{x}, \{\mathbf{W}_l\})$  represents the residual mapping to be learned (comprising convolutions, batch normalization, and ReLU activations) for the  $l$ -th layer, the output  $\mathbf{y}$  of the residual block is defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_l\}) + \mathbf{x}. \quad (8)$$

In the context of magnetic domain images, these skip connections allow the network to simultaneously preserve low-level spatial features (like abrupt chiral boundaries) while composing deep, high-level abstractions of the global thermodynamic state (see Figure 2).

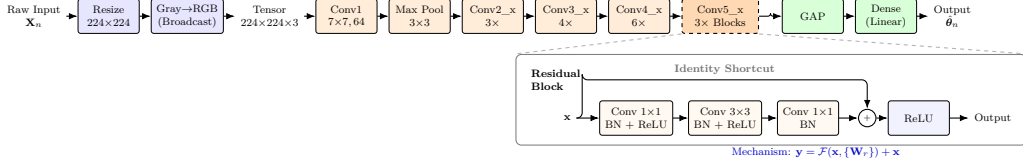


Figure 2: ResNet50 architecture pipeline for Hamiltonian parameter estimation. The pipeline progresses from single-channel input preprocessing through a pre-trained feature extractor. The Residual bottleneck structure used extensively in Conv5\_x. The deepest features are processed by the customized regression head containing a Global Average Pooling (GAP) and a final dense layer with linear activation to output the predicted physical constants.

- *Densely Connected Convolutional Networks (DenseNet)* [58]: While ResNet adds features via summation, DenseNet concatenates them, maximizing information flow and feature reuse throughout the network. This is particularly advantageous for the Hamiltonian inverse problem, as the delicate energetic competition often manifests in subtle, overlapping spatial frequencies that require dense multi-scale analysis. In a DenseNet block, the feature maps of all preceding layers are used as inputs to the current layer. If  $\mathbf{x}_l$  denotes the output of the  $l$ -th layer, and  $H_l(\cdot)$  represents a composite function of batch normalization, ReLU, and convolution, the forward pass is expressed as:

$$\mathbf{x}_l = H_l([\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{l-1}]), \quad (9)$$

where  $[\dots]$  denotes the concatenation operation. This dense topological structure ensures that the final regression head has direct access to both fine-grained pixel interactions and coarse global structures (see Figure 3).

- *Xception (Extreme Inception)* [59]: This architecture operates on the hypothesis that cross-channel correlations and spatial correlations in feature maps can be entirely decoupled. It replaces standard Inception modules with depthwise separable convolutions, drastically reducing computational overhead while enhancing representational efficiency. A depthwise separable convolution splits the standard convolution into two distinct steps. First, a depthwise convolution applies a single spatial filter to each input channel independently. Second, a pointwise convolution ( $1 \times 1$  convolution) projects the output of the depthwise



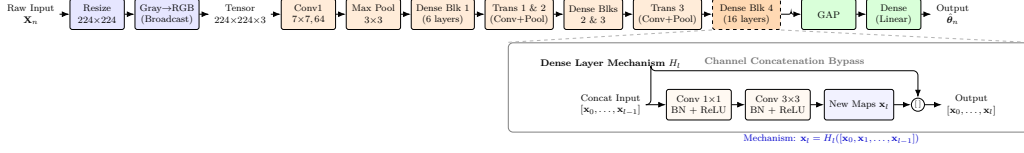


Figure 3: DenseNet121 architecture pipeline for Hamiltonian parameter estimation. The pipeline progresses left-to-right, processing single-channel inputs through the feature extractor composed of Dense Blocks and Transition layers. A detailed horizontal view illustrates the dense connectivity mechanism where feature maps from all preceding layers are concatenated before convolution. The final aggregated features are mapped to physical constants via the customized regression head.

step onto a new channel space to capture cross-channel correlations. If  $\mathbf{X} \in \mathbb{R}^{\check{H} \times \check{W} \times \check{C}}$  is the input tensor, the operation is formally approximated as:

$$\text{Conv}_{\text{sep}}(\mathbf{X}) = \text{Conv}_{\text{pointwise}}(\text{Conv}_{\text{depthwise}}(\mathbf{X})). \quad (10)$$

For nanoscale magnetism, this extreme decoupling allows the network to efficiently isolate specific topological frequency patterns (depthwise) before correlating them to the complex parameter space  $\theta_n$  (see Figure 4).

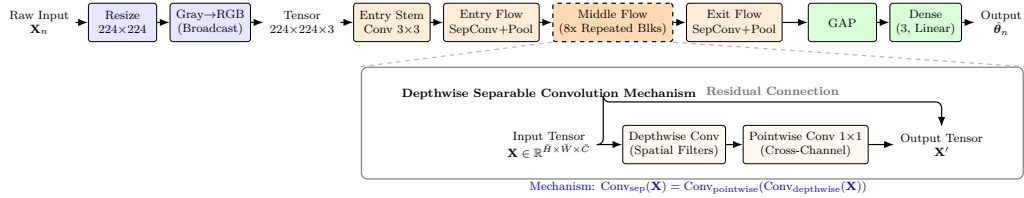


Figure 4: Xception architecture pipeline for Hamiltonian parameter estimation. The feature extractor composed of Entry, Middle, and Exit flows. A detailed horizontal view illustrates the depthwise separable convolution mechanism where spatial feature extraction (depthwise) and cross-channel correlations (pointwise) are entirely decoupled. The final aggregated features are mapped to physical constants.

- *Visual Transformer (ViT)* [60]: Unlike Convolutional Neural Networks, which enforce a strict local inductive bias, the Vision Transformer processes the 2D spatial magnetization field  $\mathbf{X}_n$  as a sequence of flattened, non-overlapping 2D patches. This allows the model to capture long-range, global dependencies across the nanodot instantly, which is criti-

cal for understanding macro-scale topological stability dictated by the Zeeman field or global anisotropy. The input image is divided into  $N_p$  patches, which are linearly projected into a sequence of  $D$ -dimensional embeddings. The core mechanism is the Multi-Head Self-Attention (MSA), which computes attention scores to weigh the importance of different patches relative to one another. For a given set of queries  $Q$ , keys  $K$ , and values  $V$  derived from the patch embeddings, the scaled dot-product attention is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^\top}{\sqrt{d_k}} \right) V, \quad (11)$$

where  $d_k$  is the scaling factor (the dimension of the keys). By employing MSA, the ViT can explicitly correlate a localized structural defect on one edge of the nanodot with a compensatory chiral boundary on the opposite edge, mapping these distant correlations directly to the global DMI or symmetric exchange strengths.

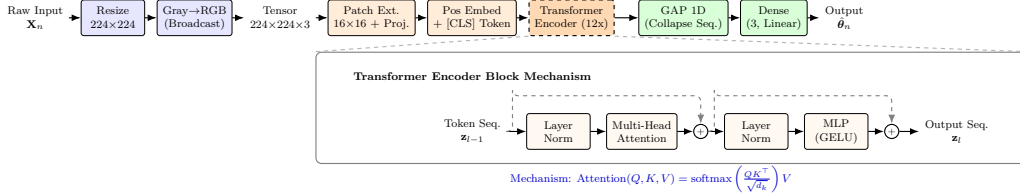


Figure 5: ViT architecture pipeline for Hamiltonian parameter estimation. The pipeline progresses left-to-right, processing single-channel inputs through patch extraction, linear projection, and a sequential backbone of Transformer Encoder blocks. The core Transformer block mechanism, highlighting the MSA that explicitly models long-range spatial dependencies across the magnetic nanodot. The final sequence of high-level token representations is aggregated via 1D GAP and mapped to the physical constants through the customized regression head.

To optimize the parameter set  $\hat{\Theta}$  in Equation 7 across all evaluated architectures, the models are trained utilizing the MSE as the explicit continuous loss function  $\mathcal{L}$ . The optimization process is executed using the Adam optimizer, relying on backpropagation and automatic differentiation to efficiently compute the network gradients [19]. By iteratively updating the model weights based on these computed gradients, the training pipeline systematically minimizes the structural and energetic discrepancies between the predicted outputs and the true

578 Hamiltonian constants, ensuring robust convergence in estimating the  
 579 underlying physical parameters.

### 580 3.4. Spatial Interpretability via RAMs

581 Interpretability methods based on activation mapping have become a  
 582 foundational approach for understanding the decision-making processes of  
 583 deep neural networks. By identifying the spatial regions within an input  
 584 that most significantly influence a model’s prediction, these methods project  
 585 opaque latent representations back into a human-interpretable domain. Among  
 586 these, Gradient-weighted Class Activation Mapping (Grad-CAM) and its  
 587 variants have been extensively utilized in computer vision to generate class-  
 588 specific saliency maps [61].

589 In standard classification paradigms, given an input image  $\mathbf{X}$  and a discrete  
 590 target class score  $y^c$ , activation maps are computed by extracting the  
 591 gradients of  $y^c$  with respect to the  $k$ -th feature map  $\mathbf{A}^k \in \mathbb{R}^{H' \times W'}$  of a terminal  
 592 convolutional layer (where  $H'$  and  $W'$  denote the spatial dimensions  
 593 of the feature map). The spatial importance weights are derived by globally  
 594 pooling these gradients, and the final map is a ReLU-activated linear  
 595 combination of the feature maps [62]. While highly effective for discrete  
 596 semantic categorization, this standard formulation structurally fails in continuous  
 597 regression tasks. The absence of a discrete target class removes the  
 598 natural selection mechanism required to isolate the output dimension driving  
 599 the interpretability process. Furthermore, directly computing gradients  
 600 with respect to a prediction  $\hat{\boldsymbol{\theta}}$  introduces severe gradient instability; minute  
 601 fluctuations in the scalar output can yield drastically different spatial attributions,  
 602 failing to provide the physical consistency required for analyzing  
 603 complex Hamiltonian parameters [63].

#### 604 3.4.1. RAMs for CNN-based Approaches

605 To overcome the fundamental limitations of discrete activation methods  
 606 in highly degenerate physical systems, we introduce RAMs, a novel formulation  
 607 explicitly tailored to bridge continuous parameter estimation with spatial  
 608 topologies. Given an input magnetic domain visualization  $\mathbf{X}$  and the  
 609 optimized regression model  $f_{\hat{\boldsymbol{\theta}}} : \mathbb{R}^{\tilde{H} \times \tilde{W}} \rightarrow \mathbb{R}^{\tilde{P}}$ , the network predicts a vector  
 610 of physical constants  $\hat{\boldsymbol{\theta}} = f_{\hat{\boldsymbol{\theta}}}(\mathbf{X})$ . Let  $\hat{\theta}_p \in \hat{\boldsymbol{\theta}}$  denote the continuous prediction  
 611 for the  $p$ -th target Hamiltonian parameter (e.g., the DMI strength  
 612  $\tilde{D}$  or symmetric exchange  $\tilde{J}$ ), and let  $\theta_p \in \boldsymbol{\theta}$  represents its corresponding  
 613 ground truth. Instead of utilizing a raw predicted scalar, we formulate a

614 scale-invariant, error-based activation objective  $\mathcal{O}_p$ . To prevent the math-  
615 ematical discontinuities and negative objective bounds inherent to absolute  
616 error formulations  $(1 - |\theta_p - \hat{\theta}_p|)$ , we utilize a normalized Gaussian decay  
617 function:

$$\mathcal{O}_p = \exp \left( -\gamma_p \left( \theta_p - \hat{\theta}_p \right)^2 \right), \quad (12)$$

618 where  $\gamma_p > 0$  is a hyperparameter-specific precision factor determining the  
619 strictness of the error decay. This objective smoothly bounds the activation  
620 target strictly within  $(0, 1]$ , promoting maximum gradient response precisely  
621 when the network’s prediction asymptotically approaches the true thermo-  
622 dynamic value, thereby isolating the topological features that positively con-  
623 tribute to accurate regression.

624 The spatial importance weights  $\alpha_k^{(p)}$  for each continuous feature map  $\mathbf{A}^k$   
625 are subsequently computed by globally averaging the gradients of the error  
626 objective  $\mathcal{O}_p$  with respect to the local spatial activations  $\mathbf{A}_{ij}^k$ :

$$\alpha_k^{(p)} = \frac{1}{H'W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} \frac{\partial \mathcal{O}_p}{\partial \mathbf{A}_{ij}^k}. \quad (13)$$

627 These weights strictly quantify the importance of the  $k$ -th latent spatial  
628 feature in successfully minimizing the prediction error for the target physical  
629 parameter  $\theta_p$ . The resulting RAM is then formulated as:

$$\mathcal{R}_{\text{RAM}}^{(p)} = \text{ReLU} \left( \sum_k \alpha_k^{(p)} \mathbf{A}^k \right). \quad (14)$$

630 The application of the ReLU activation ensures that only spatial features  
631 exerting a positive, stabilizing influence on the estimation of the target pa-  
632 rameter are preserved. Finally,  $\mathcal{R}_{\text{RAM}}^{(p)}$  is spatially upsampled to the original  
633 input resolution  $\check{H} \times \check{W}$  and normalized to the interval  $[0, 1]$ . Then, by map-  
634 ping the continuous error gradients back to the input spatial magnetization  
635 field  $\mathbf{X}$ , RAMs allow for the direct visual and quantitative identification of  
636 the specific emergent textures that the network utilizes to deduce the un-  
637 derlying energetic balances. The complete workflow for this localized spatial  
638 interpretability is illustrated in Figure 6.

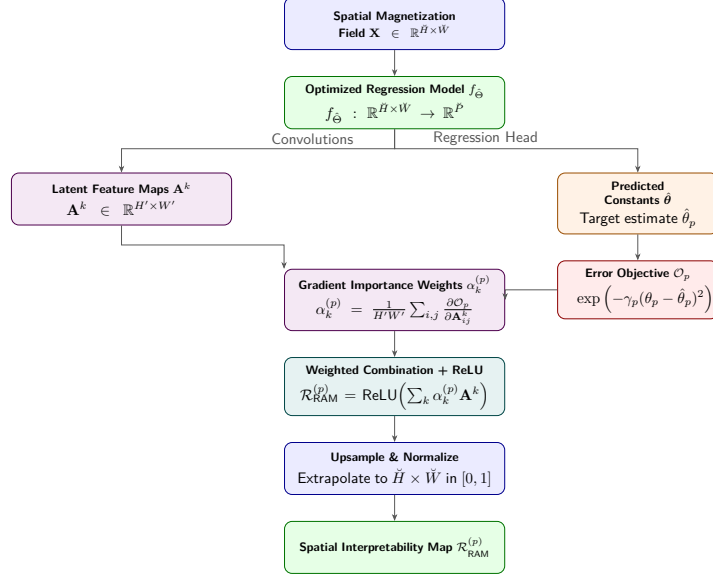


Figure 6: RAMs extraction pipeline. Latent feature maps and continuous predictions from the network backbone are explicitly coupled via an error-decay objective. Gradient-based importance weights combine these features to produce a normalized, physically traceable spatial saliency map for the target Hamiltonian parameter  $\theta_p$ .

### 3.4.2. RAMs for ViT-based Approaches

While our standard RAMs formulation seamlessly integrates with CNNs by exploiting their innate spatial hierarchies, extending this interpretability metric to the Vision Transformers requires mapping sequence-based latent representations back to the two-dimensional physical domain. Unlike CNNs, which preserve spatial locality through sliding convolutional filters, the ViT processes the input spatial magnetization field  $\mathbf{X}$  as a flattened sequence of  $N_r$  non-overlapping patches, outputting a discrete sequence of latent embeddings. To evaluate ViT-based architectures within the exact same RAMs framework, we introduce a token-reshaping mechanism that bridges the one-dimensional transformer latent space with the two-dimensional topological geometry. Let the output of the terminal Transformer encoder block be denoted as  $\mathbf{Z} \in \mathbb{R}^{(N_r+1) \times D}$ , where  $D$  is the embedding dimension and the extra token corresponds to the global aggregation token (e.g., the [CLS] token) utilized by the regression head to predict the Hamiltonian parameters  $\hat{\mathbf{y}}$ . By isolating the  $N_r$  spatial tokens, we obtain the localized sequence  $\mathbf{Z}_s \in \mathbb{R}^{N_r \times D}$ .

Because each sequence token explicitly corresponds to a fixed physical patch of the original nanodot domain,  $\mathbf{Z}_s$  can be folded back into a 2D spatial grid. Assuming square patches, the sequence is reshaped into a pseudo-feature map:

$$\mathbf{A}_{\text{ViT}} = \text{Reshape}(\mathbf{Z}_s) \in \mathbb{R}^{\sqrt{N_r} \times \sqrt{N_r} \times D}. \quad (15)$$

Once projected into this tensor format,  $\mathbf{A}_{\text{ViT}}$  functions structurally identically to the terminal convolutional feature map  $\mathbf{A}$  utilized in the CNN baseline. The  $k$ -th slice of the embedding dimension ( $k \in \{1, \dots, D\}$ ) forms the latent spatial map  $\mathbf{A}_{\text{ViT}}^k \in \mathbb{R}^{\sqrt{N_r} \times \sqrt{N_r}}$ . We then apply the identical scale-invariant, error-based activation objective  $\mathcal{O}_p$  defined in Equation 12. The spatial importance weights for the ViT embeddings are computed by globally averaging the gradients of this objective with respect to the reshaped spatial tokens:

$$\alpha_k^{(p)} = \frac{1}{N_r} \sum_{i=1}^{\sqrt{N_r}} \sum_{j=1}^{\sqrt{N_r}} \frac{\partial \mathcal{O}_p}{\partial (\mathbf{A}_{\text{ViT}}^k)_{ij}}. \quad (16)$$

RAMs for the ViT are constructed via the ReLU-activated linear combination of these reshaped embedding slices, precisely mirroring Equation 14, before being upsampled to the original input resolution  $\check{H} \times \check{W}$ . This extension guarantees an architecture-agnostic evaluation of interpretability, ensuring that the isolated degenerate textures driving the model’s predictions can be fairly compared across fundamentally disparate deep learning architectures.

### 3.4.3. Normalized RAMs for Layer-Wise and Target-Specific Analysis

To rigorously compare the spatial activations across different Hamiltonian parameters and network depths, it is imperative to compute the RAMs utilizing strictly normalized predictions and targets. Let  $\bar{\theta}_p \in [0, 1]$  and  $\hat{\theta}_p \in [0, 1]$  denote the Min-Max normalized ground truth and predicted values for the  $p$ -th physical parameter, respectively. By scaling the variables to a uniform interval, the layer-specific error objective is strictly bounded and evaluated as  $\mathcal{O}_p = \exp(-\gamma_p(\bar{\theta}_p - \hat{\theta}_p)^2)$ . For a given convolutional or transformer layer  $l \in \{1, \dots, L\}$ , we extract the layer-specific unnormalized RAM, denoted as  $\mathcal{R}_{\text{RAM}}^{(l,p)}$ , following the gradient and activation protocols established in Equations 13 and 14. To quantitatively evaluate and isolate the relative

685 contribution of each distinct layer to the estimation of every physical con-  
 686 stant, we introduce a global cross-layer and cross-parameter normalization  
 687 scheme. The globally normalized activation map,  $\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)} \in [0, 1]$ , is defined  
 688 by scaling the layer-specific maps against the absolute maximum spatial ac-  
 689 tivation observed across all  $L$  layers and all  $\check{P}$  parameters:

$$\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)} = \frac{\mathcal{R}_{\text{RAM}}^{(l,p)}}{\max_{l' \in \{1, \dots, L\}, p' \in \{1, \dots, \check{P}\}} \left( \max_{i,j} \left[ \mathcal{R}_{\text{RAM}}^{(l',p')} \right]_{i,j} \right)}. \quad (17)$$

690 This comprehensive normalization ensures that the gradient magnitudes  
 691 are dimensionally consistent and comparable, preventing parameters with  
 692 intrinsically higher variance or layers with larger gradient flows from dispropor-  
 693 tionately dominating the interpretability metrics. By tracking the evolu-  
 694 tion of  $\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)}$  across the network depth, this formulation explicitly reveals  
 695 how the deep learning architecture bypasses the intrinsically ill-conditioned  
 696 nature of the inverse problem. In magnetic nanodots, severe structural ambi-  
 697 guity arises because disparate energetic combinations of the symmetric ex-  
 698 change ( $\tilde{J}$ ), DMI ( $\tilde{D}$ ), magnetocrystalline anisotropy ( $\tilde{K}$ ), and Zeeman field  
 699 ( $\tilde{H}$ ) can yield highly degenerate states with nearly identical free energies.  
 700 Then, the global normalization aim to code how the network dynamically  
 701 resolves this degeneracy through a hierarchical physical decoupling. We es-  
 702 tablish that early network layers (lower  $l$ ) exhibit high relative activations for  
 703 parameters governing short-range spin interactions, capturing the localized  
 704 chiral winding angles and domain wall boundaries dictated primarily by the  
 705 delicate competition between  $\tilde{J}$  and  $\tilde{D}$ . Conversely, deeper layers (higher  $l$ )  
 706 aggregate these localized topological textures to resolve macro-scale thermo-  
 707 dynamic stability. These terminal layers exhibit dominant spatial activations  
 708 for  $\tilde{H}$  and  $\tilde{K}$ , which govern the global topological confinement, phase tran-  
 709 sitions, and spatial alignment of emergent structures such as skyrmion cores  
 710 and labyrinthine patterns. By correlating the intensity and spatial local-  
 711 ization of  $\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)}$  across the network hierarchy, our framework utilizes both  
 712 localized boundary chirality and global domain spacing to disambiguate the  
 713 underlying Hamiltonian parameters.

## 714 4. Experimental Set-Up

715 To evaluate the effectiveness of the proposed framework for Hamilto-  
716 nian parameter estimation, we design a comprehensive experimental protocol  
717 that integrates physics-based simulations, data-driven modeling, and inter-  
718 pretability analysis. The objective of this setup is to systematically assess not  
719 only the predictive performance of the model, but also its ability to generalize  
720 across different physical regimes and provide meaningful spatial explanations  
721 through the proposed RAMs. The dataset utilized in this study consists en-  
722 tirely of simulated magnetic domain configurations generated through our  
723 atomistic Monte Carlo framework, as described in Section 3.2. To accurately  
724 reflect the finite-size effects and surface boundaries inherent to experimental  
725 nanostructures, the magnetic nanodots were modeled on a three-dimensional  
726 discrete lattice. The cylindrical geometry was constrained by a radius of  
727  $R_d = 18.25$  [MUC] and a thickness of  $\tilde{L} = 5$  [MUC]. Within this confined  
728 grid, the spatial bounds were dictated by a strictly defined circular binary  
729 mask, yielding an active interacting volume of approximately 5,200 localized  
730 spins per simulated dot.

731 For each unique combination of Hamiltonian parameters, the system was  
732 subjected to a rigorous simulated annealing protocol to isolate the ther-  
733 modynamically relaxed state. To guarantee full ergodicity, the exchange-  
734 normalized initial temperature was established at an effective paramagnetic  
735 ceiling. The system was systematically cooled to a near-ground state of  
736  $T^{(f)} = 0.1$  K utilizing a linear decrement of  $\Delta T = -0.1$  K. At each dis-  
737 crete temperature interval, the spin lattice underwent 8,000 Monte Carlo  
738 Sweeps (MCS) exclusively dedicated to thermalization, thereby overcoming  
739 metastable localized energetic barriers. This thermalization phase was sub-  
740 sequently followed by 100 measurement MCS to compute macroscopic ther-  
741 modynamic averages (e.g., specific heat and magnetic susceptibility) and ver-  
742 ify strict energetic convergence. To overcome the prohibitive computational  
743 bottleneck traditionally associated with sequential Monte Carlo methods,  
744 the atomistic simulation was rigorously optimized utilizing PyTorch tensor  
745 operations on CUDA-enabled GPUs.

746 Upon converging to a [low-energy thermodynamically stable state](#) at 0.1 K,  
747 the final topological state was isolated. To mimic experimental 2D observa-  
748 tion techniques, the spatial magnetization field was extracted exclusively  
749 from the central discrete layer of the nanodot ( $z = \lfloor \tilde{L}/2 \rfloor$ ). The out-of-plane  
750 spin components ( $s_z \in [-1, 1]$ ) were systematically normalized to the strict



interval  $[0, 1]$  and mapped onto a Cartesian pixel grid, finalizing the input images  $\mathbf{X}$  corresponding to the specific physical targets  $\boldsymbol{\theta}$ . The extended Hamiltonian parameter vector  $\boldsymbol{\theta} \in \mathbb{R}^8$  estimated in this study is defined as:  $\boldsymbol{\theta} = [T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}]$ , where the first-shell exchange constant  $\tilde{J}_1 = 1.0$  meV is fixed as the energy reference scale to break the scale symmetry of the inverse problem and ensure a well-posed regression target, as described in Section 3.1. Each of the eight free parameters was independently and uniformly sampled within the physically motivated bounds summarized in Table 1.

It is important to note that while all eight parameters are simultaneously estimated by the regression models, their effective identifiability from the  $s_z$  projection varies substantially due to the physical constraints of the observational modality. The higher-order exchange constants  $\tilde{J}_3$  and  $\tilde{J}_4$  produce energetic contributions that are strongly dominated by the first- and second-shell interactions  $\tilde{J}_1$  and  $\tilde{J}_2$ , rendering their individual spatial signatures largely indistinguishable in the magnetization texture. Similarly, the magnetocrystalline anisotropy constants  $\tilde{K}_{an1}$  and  $\tilde{K}_{anS}$  and the external field  $\tilde{H}_{ex}$  couple primarily to the in-plane spin components  $s_x$  and  $s_y$ , which are not observable in the  $s_z$  projection extracted from the central layer. These parameters were nevertheless included in the regression target  $\boldsymbol{\theta}$  to provide a complete and physically honest characterization of the inverse problem, including its fundamental identifiability limits. Their inclusion does not degrade the predictive accuracy of the three most identifiable parameters ( $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{K}_{DM}$ ), since the multi-output regression head learns to allocate gradient capacity independently to each output dimension. The resulting dataset is publicly available and can be accessed through the Kaggle repository<sup>1</sup>. It contains a total of 218,256 samples, where each input is a grayscale magnetic domain image of size  $39 \times 39 \times 1$  pixels, and each corresponding target is a vector of eight physical parameters  $\boldsymbol{\theta} \in \mathbb{R}^8$  as defined above.

It is important to note that the DMI strength  $\tilde{K}_{DM}$  is sampled exclusively over non-negative values ( $\tilde{K}_{DM} \in [0.0, 1.2]$  meV). This restriction is a physically motivated convention arising from the symmetry properties of the antisymmetric exchange term in Equation ???. Under the transformation  $\tilde{D} \rightarrow -\tilde{D}$ , the equilibrium spin configuration undergoes a reversal of

---

<sup>1</sup><https://www.kaggle.com/datasets/carlosanamejoy/dataset-spines-complete>

Table 1: Sampling ranges for the eight free Hamiltonian parameters constituting the regression target  $\theta$ . The first-shell exchange  $\tilde{J}_1 = 1.0$  meV is fixed as the energy reference. Parameters are sampled independently and uniformly within the specified bounds. Physical units: temperature in Kelvin [K], exchange and DMI constants in meV, anisotropy constants in meV/atom, external field in meV/atom.

| Parameter                    | Physical role         | Min   | Max  |
|------------------------------|-----------------------|-------|------|
| $T^{(0)}$ [K]                | Annealing temperature | 0.2   | 20.0 |
| $\tilde{J}_2$ [meV]          | 2nd-shell exchange    | −0.66 | 0.66 |
| $\tilde{J}_3$ [meV]          | 3rd-shell exchange    | −0.29 | 0.29 |
| $\tilde{J}_4$ [meV]          | 4th-shell exchange    | −0.23 | 0.23 |
| $\tilde{K}_{an1}$ [meV/atom] | Bulk anisotropy       | 0.0   | 0.2  |
| $\tilde{K}_{anS}$ [meV/atom] | Surface anisotropy    | 0.0   | 0.2  |
| $\tilde{H}_{ex}$ [meV/atom]  | External Zeeman field | 0.0   | 0.05 |
| $\tilde{K}_{DM}$ [meV]       | DMI strength          | 0.0   | 1.2  |

its in-plane chirality—the rotational sense of the chiral winding inverts—while the out-of-plane magnetization component  $s_z$  at every lattice site remains strictly invariant. Since the regression models operate exclusively on the two-dimensional  $s_z$  projection, configurations with  $+\tilde{D}$  and  $-\tilde{D}$  are exactly degenerate in the observation space, rendering the sign of the DMI fundamentally unidentifiable from  $s_z$  images alone. Including both signs simultaneously would introduce a strict two-fold degeneracy in the input–output mapping, collapsing the model’s predictions toward  $\hat{\tilde{D}} \approx 0$  and precluding meaningful regression. The chirality handedness is instead encoded through the fixed direction vector  $\hat{\mathbf{d}}_{ij}$ , consistent with standard conventions in atomistic modeling of interfacial DMI [53? ]. Recovering the DMI sign would require access to chirality-sensitive observables, such as the full three-dimensional spin vector field or in-plane magnetization components, which constitutes a direction for future investigation (Section 6).

To ensure robust evaluation and prevent information leakage during the deep learning optimization phase, the generated dataset  $\mathcal{D}$  is partitioned into training, validation, and testing subsets using a rigorous two-stage random split strategy. Initially, 15% of the total dataset is isolated as a hold-out test set to evaluate ultimate model generalization. Subsequently, the remaining 85% is further divided, allocating exactly 15% of the total initial data for validation. This procedure yields an effective split of 70% for training, 15% for validation, and 15% for testing. All partitions are executed utilizing a

fixed random seed to guarantee strict reproducibility across all experimental runs. This strategy ensures that topological samples remain strictly independent across subsets while perfectly preserving the underlying thermodynamic distributions of the dataset.

Prior to network ingestion, both the input spatial magnetization images and the target Hamiltonian variables are preprocessed to ensure numerical stability and efficient gradient convergence. The physical target variables are scaled using Min-Max normalization, mapping each continuous parameter to the strict interval  $[0, 1]$  based exclusively on the statistical distribution of the training set. This identical transformation is subsequently applied to the validation and test sets to maintain inferential consistency and directly support the bounded RAMs objective  $\mathcal{O}_p \in (0, 1]$ . For the input images ( $\mathbf{X}_n$ ), the single-channel  $s_z$  matrices are spatially resized to  $224 \times 224$  pixels and duplicated across three channels (grayscale-to-RGB conversion). This dimensional expansion is critical to seamlessly leverage the hierarchical spatial filters learned from an ImageNet-based pretraining [64].

The configuration of the evaluated deep learning architectures (see Section 3.3) is summarized in Table 2. All four architectures share a unified preprocessing pipeline and an extended regression head designed to mitigate overfitting in the high-dimensional parameter space. Specifically, the global pooling output is passed through a Batch Normalization layer, followed by a Dropout layer with rate  $p = 0.4$ , a fully connected Dense layer of 256 units with ReLU activation and  $\ell_2$  weight regularization ( $\lambda = 10^{-4}$ ), a second Batch Normalization layer, a second Dropout layer with rate  $p = 0.3$ , and a final linear Dense layer outputting the  $\tilde{P} = 8$  target parameters. For the CNN-based architectures (ResNet50, DenseNet121, and Xception), the global aggregation is performed via Global Average Pooling over the spatial feature maps, whereas for the ViT-B/16 the token sequence output of the terminal encoder block is first collapsed via Global Average Pooling along the sequence dimension before entering the shared regularization head. This design choice reflects the distinct spatial inductive biases of convolutional and attention-based backbones while ensuring a strictly comparable regression capacity across all evaluated models.

The optimization of the parameter set  $\hat{\Theta}$  was executed utilizing the Adam optimizer with a unified initial learning rate of  $\eta = 10^{-4}$  for all architectures, including the ViT-B/16, reflecting the well-established sensitivity of both transformer-based and deep convolutional models to aggressive learning rate schedules when fine-tuning from ImageNet pretraining. All models

Table 2: Configuration of evaluated deep learning architectures for inverse Hamiltonian parameter estimation. All models share a standardized regularized regression head mapped to the  $\dot{P}$  = 8-dimensional physical parameter space. BN: Batch Normalization; Drop( $p$ ): Dropout with rate  $p$ .

| Model       | Architecture Type       | Pretraining | Global Aggregation            | Output Head                                                                                                                          |
|-------------|-------------------------|-------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ResNet50    | Residual CNN            | ImageNet    | Global Avg. Pooling (spatial) | BN $\rightarrow$ Drop(0.4) $\rightarrow$ Dense(256, ReLU, $\ell_2$ ) $\rightarrow$ BN $\rightarrow$ Drop(0.3) $\rightarrow$ Dense(8) |
| DenseNet121 | Densely Connected CNN   | ImageNet    | Global Avg. Pooling (spatial) | BN $\rightarrow$ Drop(0.4) $\rightarrow$ Dense(256, ReLU, $\ell_2$ ) $\rightarrow$ BN $\rightarrow$ Drop(0.3) $\rightarrow$ Dense(8) |
| Xception    | Depthwise Separable CNN | ImageNet    | Global Avg. Pooling (spatial) | BN $\rightarrow$ Drop(0.4) $\rightarrow$ Dense(256, ReLU, $\ell_2$ ) $\rightarrow$ BN $\rightarrow$ Drop(0.3) $\rightarrow$ Dense(8) |
| ViT-B/16    | Patch-based Transformer | ImageNet    | Global Avg. Pooling (tokens)  | BN $\rightarrow$ Drop(0.4) $\rightarrow$ Dense(256, ReLU, $\ell_2$ ) $\rightarrow$ BN $\rightarrow$ Drop(0.3) $\rightarrow$ Dense(8) |

were trained to minimize the MSE loss function, while the MAE was actively monitored as a secondary diagnostic. Training was conducted using a highly parallelized batch size of 512 samples over a maximum of 100 epochs. To actively prevent overfitting and manage the learning dynamics within the highly degenerate optimization landscape, two primary callbacks were implemented: a dynamic learning rate scheduler (`ReduceLROnPlateau`) that reduced the learning rate by a factor of 0.3 if the validation loss stagnated for 4 consecutive epochs, and an `EarlyStopping` protocol that halted training and restored the best network weights if no validation improvement was observed after 8 consecutive epochs.

The predictive performance on the unseen test set was quantitatively assessed using two complementary regression metrics. The coefficient of determination ( $R^2$ ) quantifies the proportion of variance in each physical parameter successfully captured by the network:

$$R^2 = 1 - \frac{\sum_{n=1}^{N_{\text{test}}} (\theta_{n,p} - \hat{\theta}_{n,p})^2}{\sum_{n=1}^{N_{\text{test}}} (\theta_{n,p} - \bar{\theta}_p)^2}, \quad (18)$$

where  $\bar{\theta}_p$  is the mean of the observed ground truth target. Note that  $R^2$  is statistically undefined when the target variable exhibits near-zero variance within a given evaluation subset; in these cases the metric is reported as N/A. The Mean Absolute Error (MAE) evaluates the absolute magnitude of the prediction error in the normalized feature space  $[0, 1]$ , providing a direct and interpretable measure of per-parameter accuracy:

$$\text{MAE} = \frac{1}{N_{\text{test}}} \sum_{n=1}^{N_{\text{test}}} |\theta_{n,p} - \hat{\theta}_{n,p}|. \quad (19)$$

Since all target variables are Min-Max normalized to  $[0, 1]$  prior to training, the MAE is inherently dimensionless and directly comparable across all

867 eight Hamiltonian parameters, irrespective of their original physical units or  
 868 numerical ranges.

869 The RAMs were computed on the held-out test set using the formulations  
 870 introduced in Section 3.4. For the Xception backbone, RAMs were extracted  
 871 from eight layers selected to systematically cover the three functional stages  
 872 of the architecture: two entry-flow layers capturing low-level spatial features  
 873 at high resolution — `block1_conv2` ( $112 \times 112$ ) and `block3_sepconv2` ( $56 \times$   
 874  $56$ ) —; one transitional layer at `block4_sepconv2` ( $28 \times 28$ ); five middle-flow  
 875 layers operating at the bottleneck resolution of  $14 \times 14$  — `block6_sepconv2`,  
 876 `block8_sepconv2`, `block10_sepconv2`, and `block12_sepconv2` —; and one  
 877 exit-flow layer, `block13_sepconv2` ( $14 \times 14$ ), immediately preceding the  
 878 global pooling operation. This stratified selection was motivated by the  
 879 physical hypothesis that early layers encode fine-grained domain periodic-  
 880 ity and local spin-winding patterns at the scale of individual domain walls,  
 881 while middle-flow layers progressively aggregate these local features into in-  
 882 termediate topological representations, and exit-flow layers encode high-level  
 883 semantic information about the global magnetic phase. By distributing the  
 884 selected layers across all three functional stages, the analysis directly tests  
 885 whether the hierarchical decoupling predicted by the normalized RAM for-  
 886 mulation of Equation 17 is empirically realized in the Xception backbone for  
 887 Hamiltonian parameter estimation. For the ViT, a single attribution map  
 888 per parameter was derived from the spatial token embeddings at the output  
 889 of the terminal encoder block, reshaped into  $\mathbf{A}_{\text{ViT}} \in \mathbb{R}^{14 \times 14 \times 768}$  following  
 890 Equation 15. Since `keras_hub.ViTBackbone` does not expose internal token  
 891 tensors as differentiable nodes within `tf.GradientTape`, the token sequence  
 892  $\mathbf{Z}_s \in \mathbb{R}^{196 \times 768}$  was first extracted via a standard forward pass and then re-  
 893 instantiated as a `tf.Variable` to enable gradient computation. The [CLS]  
 894 token was treated as a fixed constant and the reconstructed sequence was  
 895 passed through the `GlobalAveragePooling1D` and `Dense` regression head to  
 896 compute  $\mathcal{O}_p$  and its gradients with respect to  $\mathbf{Z}_s$ , following Equation 16.

897 The precision factor was fixed to  $\gamma_p = 10$  uniformly for all  $p \in \{1, \dots, 8\}$ .  
 898 Given that all target variables are Min-Max normalized to  $[0, 1]$ , this value  
 899 ensures a smooth and well-conditioned gradient signal across the full predic-  
 900 tion error range:  $\mathcal{O}_p \geq e^{-10} \approx 4.5 \times 10^{-5}$  at maximum error and  $\mathcal{O}_p = 1$   
 901 at perfect prediction. Lower values of  $\gamma_p$  would yield spatially diffuse at-  
 902 tributions insensitive to prediction quality, while higher values would cause  
 903 gradient saturation for the less identifiable parameters where errors are sys-  
 904 tematically larger. The chosen value maintains sensitivity across all eight

parameters, from the most identifiable ( $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{K}_{DM}$ ) to the physically degenerate ones ( $\tilde{K}_{an1}$ ,  $\tilde{K}_{anS}$ ,  $\tilde{H}_{ex}$ ).

To confine activations to the physical extent of the nanodot, a circular binary mask  $\mathcal{M} \in \{0, 1\}^{224 \times 224}$  — defined as a disk of radius  $r = 0.94 \times 112$  pixels centered at the image center — was applied to each RAM before normalization, setting all activations outside the disk to zero. A concentration score  $\xi \in [0, 1]$ , measuring the fraction of total activation energy within the disk, was used to select the most physically representative sample per magnetic phase: among the ten test images closest to the UMAP centroid of each phase cluster, the sample maximizing  $\xi$  averaged over all parameters and layers was chosen as the phase representative  $\mathbf{X}_n^*$ .

All experiments were conducted in a cloud-based environment using Google Colab. The computational setup was equipped with an NVIDIA H100 GPU with 97 GB of VRAM, supported by 210 GB of system RAM and 230 GB of available disk storage. The training pipeline was implemented in Python using TensorFlow, leveraging GPU acceleration for efficient model training and inference. The complete source code is publicly available at <https://github.com/deathperminut/PaperInverseProblemEstimation>, organized into three modules: `/simulation` containing the atomistic Monte Carlo Hamiltonian simulation scripts implemented in PyTorch with CUDA GPU parallelization; `/regression` containing the training, validation, and evaluation pipelines for all four deep learning architectures; and `/rams` containing the Regression Activation Maps computation and visualization code for both the Xception CNN and the Vision Transformer.

## 5. Results and Discussion

### 5.1. Thermodynamic Simulation of Magnetic Topologies

We present the main results of the proposed atomistic Hamiltonian simulation framework (see Section 3). Figure 7 delineates the empirical probability density functions—comprising histograms and kernel density estimates—for the extended parameter space  $\theta$ , capturing the foundational statistical architecture of the simulated dataset. While the higher-order exchange constants ( $\tilde{J}_2$ ,  $\tilde{J}_3$ ,  $\tilde{J}_4$ ) and the DMI ( $\tilde{K}_{DM}$ ) exhibit tight, highly symmetric distributions, macroscopic scaling variables like the initial annealing temperature ( $T^{(0)}$ ) display pronounced right-skewed behavior. This rigorous statistical characterization is crucial because these specific distributional shapes,

940 bounded ranges, and sampling heterogeneities explicitly govern the topo-  
 941 logical diversity of the generated magnetic phase space. Consequently, this  
 942 varying sampling density directly dictates the learning dynamics, optimiza-  
 943 tion stability, and predictive limits of the downstream deep regression frame-  
 944 works, particularly when navigating the highly degenerate regimes of the  
 945 extended Heisenberg Hamiltonian.

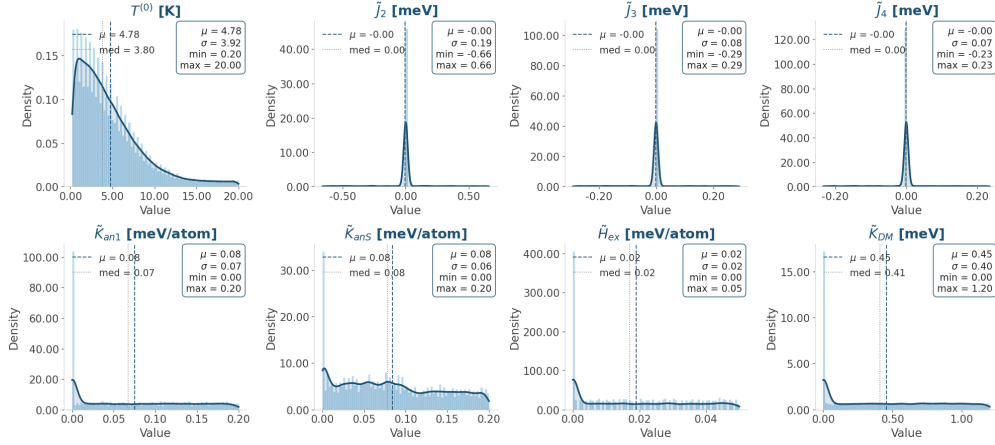


Figure 7: Empirical distributions of the thermodynamic and Hamiltonian parameters  $\{T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}\}$  used to generate the dataset. Each subplot presents a histogram and KDE for a given variable, along with summary statistics including mean, standard deviation, and value range.

946 To further validate the physical consistency of the generated dataset  
 947 and provide an intuitive understanding of the underlying spin dynamics,  
 948 Figure 8 illustrates the thermodynamic evolution of a representative mag-  
 949 netic nanodot during the Metropolis–Hastings Monte Carlo annealing pro-  
 950 cess. This simulation is conducted under the fixed parameter configura-  
 951 tion  $\tilde{J}_1 = 1.0$  meV,  $\tilde{J}_2 = \tilde{J}_3 = \tilde{J}_4 = 0.0$  meV,  $\tilde{K}_{an1} = 0.10$  meV/atom,  
 952  $\tilde{K}_{anS} = 0.00$  meV/atom,  $\tilde{H}_{ex} = 0.00$  meV/atom, and  $\tilde{K}_{DM} = 1.00$  meV,  
 953 which constitutes a representative point within the full height-dimensional  
 954 parameter space  $\boldsymbol{\theta}$  defined in Table 1. The vanishing values of the higher-  
 955 order exchange constants ( $\tilde{J}_2, \tilde{J}_3, \tilde{J}_4$ ), surface anisotropy ( $\tilde{K}_{anS}$ ), and ex-  
 956 ternal field ( $\tilde{H}_{ex}$ ) isolate the dominant energetic competition between the  
 957 first-shell symmetric exchange  $\tilde{J}_1$  and the DMI  $\tilde{K}_{DM}$ , making this configu-  
 958 ration particularly illustrative of the chiral ordering mechanism. The main  
 959 panel tracks the total Hamiltonian energy  $\mathcal{H}$  [meV] as a function of the

exchange-normalized annealing temperature  $T^{(0)}$  [K], while each inset displays the corresponding two-dimensional out-of-plane magnetization field  $s_z$  of the central layer at that specific thermal state. At elevated temperatures ( $T^{(0)} \gtrsim 15$  K), the spin configurations exhibit a fully disordered, paramagnetic character, manifesting as spatially uncorrelated fluctuations of  $s_z$  with no preferred orientation. As the system is progressively cooled through the intermediate regime ( $5 \text{ K} \lesssim T^{(0)} \lesssim 15 \text{ K}$ ), short-range correlated domains begin to nucleate, reflecting the growing dominance of the symmetric exchange interaction  $\tilde{J}_1$  over thermal fluctuations. In this regime, the competition between  $\tilde{J}_1$  and the  $\tilde{K}_{DM}$  progressively induced non-collinear and chiral spin winding, visible as the emergence of alternating red-blue stripe-like patterns. Upon convergence to the near-ground state at  $T^{(0)} = 0.1$  K, the system stabilizes into a well-defined labyrinthine domain configuration, characteristic of the helical phase governed by a strong DMI relative to the exchange energy. The monotonic decrease  $\mathcal{H}$  throughout the annealing trajectory confirms the thermodynamic relaxation of the system toward [a low-energy stable configuration](#), validating the physical correctness of the Monte Carlo pipeline and the quality of the generated dataset as input for the deep learning regression framework.

Figure 9 illustrates the converged ground-state spin configuration of a magnetic nanodot at  $T_{final}^{(0)} = 0.1$  K, simulated with specific fixed parameters ( $\tilde{J}_1 = 1.0$  meV,  $\tilde{K}_{DM} = 1.00$  meV,  $\tilde{K}_{an1} = 0.10$  meV/atom, and all other parameters zeroed; see Table 1). Panel (a) maps the 2D out-of-plane magnetization ( $s_z$ ) on the central layer—using dots for out-of-plane moments and arrows for transverse components—while Panel (b) displays the full 3D volumetric spin texture. The nanodot exhibits a labyrinthine domain pattern characterized by chiral walls. This morphology is driven directly by the energetic competition between the ferromagnetic symmetric exchange ( $\tilde{J}_1$ ) and the DMI ( $\tilde{K}_{DM}$ ). With a ratio of  $\tilde{K}_{DM}/\tilde{J}_1 = 1.0$ , the system stabilizes in the helical phase. By nullifying higher-order exchange, surface anisotropy, and external fields, this  $\tilde{J}_1$ – $\tilde{K}_{DM}$  competition is fully isolated, allowing the stripe-like domains to develop freely without preferential axial pinning.

Now, to fully characterize the structural diversity generated by the Monte Carlo annealing protocol across the entire dataset, Figure 10 presents two-dimensional phase diagrams of the magnetization image as a function of key Hamiltonian parameter pairs:  $T^{(0)}$  versus  $\tilde{J}_2$  and  $T^{(0)}$  versus  $\tilde{K}_{DM}$ . These diagrams illustrate the macroscopic topological transitions driven by the under-



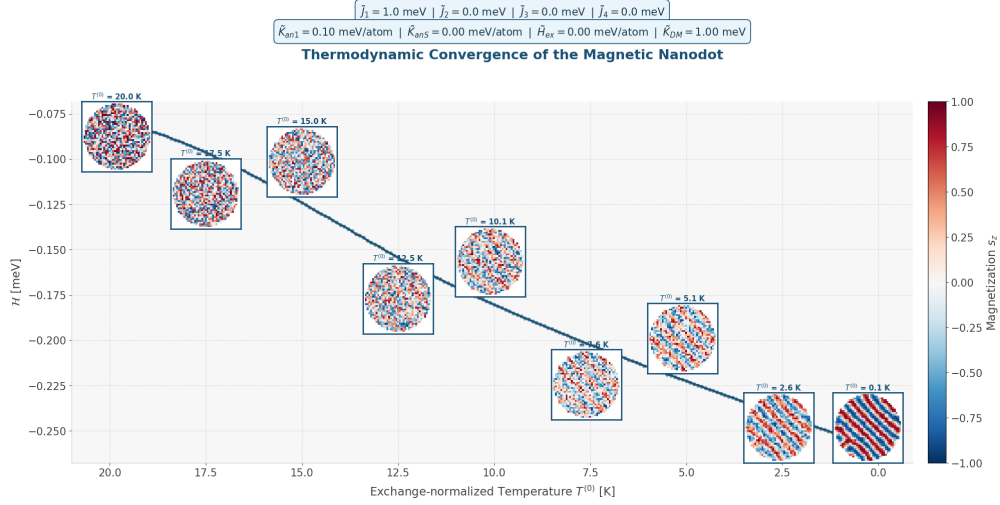


Figure 8: Thermodynamic convergence of a representative magnetic nanodot during simulated annealing. The main panel shows the total Hamiltonian energy  $\mathcal{H}$  [meV] as a function of the exchange-normalized annealing temperature  $T^{(0)}$  [K] under the fixed parameter configuration. Each inset depicts the two-dimensional out-of-plane magnetization field  $s_z$  of the central layer at the corresponding thermal state, illustrating the progressive transition from a fully disordered paramagnetic phase at high  $T^{(0)}$  to a thermodynamically relaxed labyrinthine domain configuration at  $T^{(0)} = 0.1$  K, driven by the competing energetic contributions of  $\tilde{J}_1$  and  $\tilde{K}_{DM}$ .

997 lying atomistic energetic competition. At elevated annealing temperatures,  
 998 specifically above the effective Curie temperature  $T_C$ , thermal fluctuations  
 999 overwhelm the magnetic interactions, collapsing all configurations into an  
 1000 indistinguishable, highly degenerate paramagnetic noise phase regardless of  
 1001 the specific exchange or DMI strengths. Conversely, in the low-temperature  
 1002 regime, the system stabilizes into ordered states, revealing distinct bound-  
 1003 aries between chiral, ferromagnetic, and antiferromagnetic phases. Notably,  
 1004 the phase diagrams expose critical transition zones—such as the boundary  
 1005 between chiral and disordered states near  $\tilde{K}_{DM} \approx 0$  meV, and the sign tran-  
 1006 sition near  $\tilde{J}_2 \approx 0$  meV—where disparate parameter combinations yield vi-  
 1007 sually overlapping spatial textures. By comprehensively sampling this phase  
 1008 space, the thermodynamic simulation establishes a highly diverse and struc-  
 1009 turally ambiguous dataset (ill-conditioned mapping), setting a rigorously  
 1010 challenging foundation for the subsequent inverse deep learning parameter  
 1011 estimation.

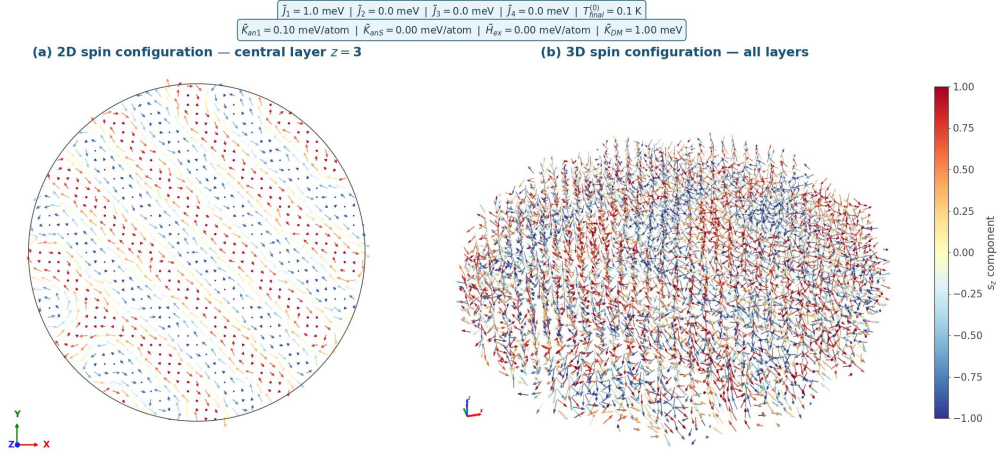


Figure 9: Converged ground-state spin configuration of a representative magnetic nanodot at  $T_{\text{final}}^{(0)} = 0.1 \text{ K}$  under a fixed parameter configuration. (a) Two-dimensional projection of the out-of-plane magnetization field  $s_z$ : colored dots indicate spins oriented predominantly out-of-plane, while colored arrows represent the in-plane orientation ( $s_x, s_y$ ) of transverse moments. (b) Full three-dimensional spin configuration, revealing the volumetric coherence of the labyrinthine chiral domain structure stabilized by the competition between  $\tilde{J}_1$  and  $\tilde{K}_{DM}$ . Color encodes the  $s_z$  component.

## 5.2. Deep Learning Regression Results

The predictive performance of the studied architectures (DenseNet, ResNet, Xception, and ViT) was assessed on the held-out test set comprising 15% of the total dataset. Each model was tasked with simultaneously inferring the complete extended parameter vector  $\theta$  directly from the two-dimensional out-of-plane magnetization image  $\mathbf{X}_n$ , without any auxiliary physical information. As shown in Figures 11 and 12, no single architecture uniformly dominates both metrics: the ViT attains the highest  $R^2$  score for seven of the eight Hamiltonian parameters ( $T^{(0)}$ ,  $\tilde{J}_3$ ,  $\tilde{J}_4$ ,  $\tilde{K}_{\text{an1}}$ ,  $\tilde{K}_{\text{anS}}$ ,  $\tilde{H}_{\text{ex}}$ , and  $\tilde{K}_{DM}$ ), while Xception achieves the highest  $R^2$  for the second-shell exchange constant  $\tilde{J}_2$ . Notably, despite the ViT's superior variance-explained performance, Xception consistently produces the lowest Mean Absolute Error across most parameters, including  $T^{(0)}$ ,  $\tilde{J}_2$ ,  $\tilde{J}_3$ ,  $\tilde{J}_4$ ,  $\tilde{H}_{\text{ex}}$ , and  $\tilde{K}_{DM}$ , whereas the ViT only minimizes the MAE for the anisotropy constants  $\tilde{K}_{\text{an1}}$  and  $\tilde{K}_{\text{anS}}$ . This complementary behavior between  $R^2$  and MAE indicates that, while the ViT captures a larger fraction of the parameter variance and is therefore more sensitive to the long-range correlations governing global mag-

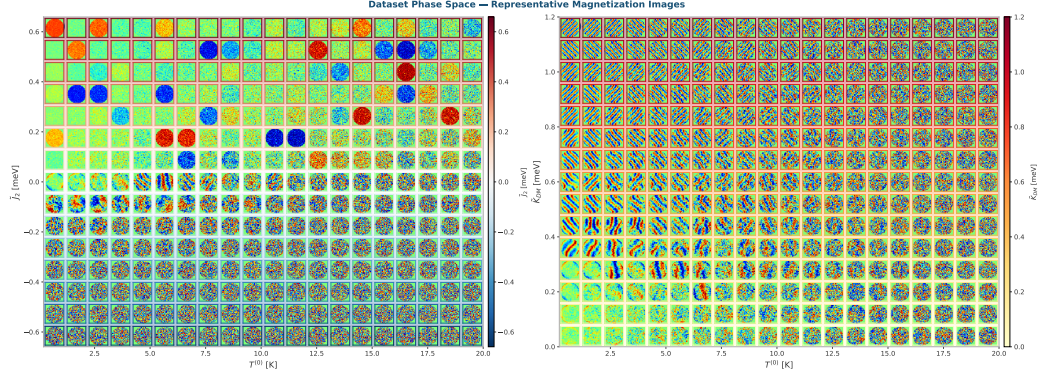


Figure 10: Two-dimensional phase diagrams of the magnetization texture  $\mathbf{X}_n$  as a function of pairs of Hamiltonian parameters. **(a)**  $T^{(0)}$  vs  $\tilde{J}_2$ . **(b)**  $T^{(0)}$  vs  $\tilde{K}_{DM}$ . Each cell displays a representative magnetization image sampled from the dataset at the corresponding parameter combination; background color encodes the value of the vertical-axis parameter.

netic order Xception yields tighter absolute predictions in the dense ordered regions of the parameter space, where local spatial structure is the dominant predictive signal. The ViT’s broad advantage in  $R^2$  is physically interpretable: by processing the magnetization field  $\mathbf{X}_n$  as a sequence of non-overlapping spatial patches and computing multi-head self-attention scores across all patch pairs, the ViT explicitly captures long-range correlations between distant spin textures, such as the global chirality imposed by  $\tilde{K}_{DM}$  and the large-scale ordering driven by  $\tilde{J}_1$ , that are fundamentally inaccessible to purely local convolutional receptive fields. Xception’s competitive behavior, in turn, stems from its depthwise separable convolution mechanism, which efficiently decouples the extraction of spatial spin-texture frequencies from the cross-channel correlations required to map those features onto the Hamiltonian parameter space, yielding the most accurate point predictions for the dominantly identifiable parameters.

Figures 11 and 12 reveal a pronounced and physically interpretable heterogeneity in predictive accuracy across the parameter space. The initial annealing temperature  $T^{(0)}$ , the second-shell exchange constant  $\tilde{J}_2$ , and the DMI strength  $\tilde{K}_{DM}$  consistently yield the highest  $R^2$  scores across all architectures, reflecting the robust and spatially distinctive topological imprints they leave on the observable magnetization texture:  $T^{(0)}$  governs the global degree of spin order,  $\tilde{J}_2$  controls the second-neighbor exchange pathway that modulates the domain-scale collinearity, and  $\tilde{K}_{DM}$  directly dictates

the characteristic wavelength and chirality of the domain patterns. However, a seemingly contradictory observation emerges when examining the MAE panels in Figure 12: despite their high  $R^2$  scores,  $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{K}_{DM}$  also exhibit comparatively larger absolute errors in their respective physical units — reaching up to approximately 1.0 K for  $T^{(0)}$ , 0.065 meV for  $\tilde{J}_2$ , and 0.05 meV for  $\tilde{K}_{DM}$  in the worst-performing architecture. This apparent inconsistency is not a failure of the models but rather a direct consequence of the thermodynamic landscape of the dataset. As illustrated in the phase diagrams of Figure 10, the parameter space contains extensive high-temperature regions where thermal fluctuations above the effective Curie temperature  $T_C$  collapse the magnetization texture into an indistinguishable paramagnetic noise phase, regardless of the specific values of  $\tilde{J}_2$  or  $\tilde{K}_{DM}$ . Within these thermally disordered regimes, visually identical configurations correspond to widely different parameter values, rendering the inverse mapping fundamentally ill-posed and inflating the absolute prediction error even when the global correlation structure — captured by  $R^2$  — remains strong. The MAE values for  $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{K}_{DM}$  therefore reflect the intrinsic degeneracy of the high-temperature paramagnetic phase rather than a systematic modeling deficiency: the same parameter that is most accurately recovered in the ordered low-temperature regime (where the spin texture preserves a clear topological signature) is simultaneously the most degenerate in the disordered high-temperature regime (where the texture collapses into noise), producing non-negligible global MAE values driven primarily by the ill-conditioned subset of the dataset. A rigorous quantitative decomposition of this effect requires first separating the dataset into its constituent magnetic structural categories, as the global aggregation of all configurations inevitably conflates high-accuracy ordered-phase predictions with fundamentally ill-posed paramagnetic-regime estimates. This per-phase analysis is addressed further ahead, where the unsupervised structural clustering of the magnetization images allows us to isolate and quantify the specific contribution of each magnetic phase to the observed prediction error.

Conversely, the models exhibit two qualitatively distinct regimes of limited predictive performance for the remaining Hamiltonian parameters, which together delimit the fundamental identifiability boundaries of the inverse problem in the 2D observation space. The higher-order exchange constants  $\tilde{J}_3$  and  $\tilde{J}_4$  constitute the most severe case: their  $R^2$  scores remain near zero across all evaluated architectures, indicating that the models are unable to recover meaningful information about these parameters beyond the dataset

mean. Physically, the spatial signatures of these subordinate exchange constants are energetically dominated by the primary  $\tilde{J}_1$  and  $\tilde{J}_2$  interactions, progressively masking their observable effect on the magnetization texture and rendering them effectively unidentifiable from the  $s_z$  projection alone. Notably, however, their MAE values — on the order of 0.03–0.05 meV — remain numerically comparable to those of the better-identified parameters, which is a direct consequence of their narrow physical range in the dataset rather than genuine predictive skill: a model predicting the dataset mean for  $\tilde{J}_3$  and  $\tilde{J}_4$  will naturally incur a small absolute error simply because the dynamic range of these parameters is itself limited. The magnetocrystalline anisotropy constants  $\tilde{K}_{an1}$  and  $\tilde{K}_{anS}$  together with the external Zeeman field  $\tilde{H}_{ex}$ , in turn, occupy an intermediate identifiability regime. This partial predictability is physically consistent: these parameters couple primarily to the in-plane spin components  $s_x$  and  $s_y$  rather than to the observable out-of-plane component  $s_z$ , so the models are able to recover them only indirectly through the secondary topological imprints they induce on the chiral domain structure, such as the modulation of stripe periodicity or the preferential orientation of skyrmion cores. Furthermore, the visual manifestations of surface anisotropy  $\tilde{K}_{anS}$  remain partially entangled with bulk anisotropy  $\tilde{K}_{an1}$  and geometrically confined to the boundary layer of the nanodot, while the Zeeman contribution  $\tilde{H}_{ex}$  — with MAE values below 0.009 meV/atom across all architectures — becomes increasingly indistinguishable from thermal disorder in the high-temperature regime. Together, these effects render the anisotropy and field parameters intrinsically partially observable and define the fundamental identifiability limits of the inverse problem within the two-dimensional  $s_z$  projection.

To gain deeper insight into the internal representations learned by the ViT during the inverse Hamiltonian estimation task, Figure 13 presents a Uniform Manifold Approximation and Projection (UMAP) of the model’s latent space [65], where each point corresponds to a test sample colored by the value of each of the eight Hamiltonian parameters  $\boldsymbol{\theta}$ . The UMAP embedding was computed from the high-dimensional feature vectors extracted from the ViT’s final encoder block prior to the regression head, providing a two-dimensional visualization of the geometric structure that the model has learned to impose on the magnetization image manifold. Rather than forming a single undifferentiated cloud, the latent space exhibits a rich topological structure composed of several geometrically distinct clusters and continuous gradients, reflecting the fact that the model has organized the magnetiza-

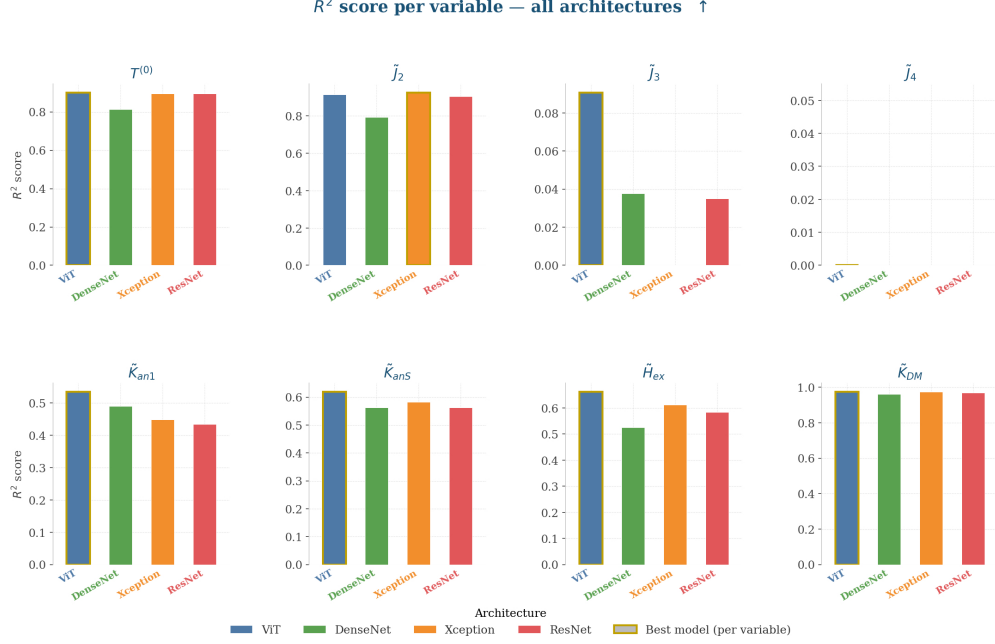


Figure 11: Per-variable  $R^2$  score across all evaluated architectures for the eight Hamiltonian parameters  $\theta$ , reported on the held-out test set. Each panel corresponds to one target parameter and shows one bar per architecture. Negative  $R^2$  values, indicating predictions no better than the dataset mean, are clipped to zero for visualization clarity. Gold-bordered bars indicate the best-performing architecture for each parameter.

tion textures according to their underlying physical regimes. Examining the coloring by  $T^{(0)}$  reveals the clearest and most spatially coherent gradient across the embedding: low-temperature ordered configurations concentrate in well-separated compact regions, while high-temperature disordered states occupy a large, diffuse cluster in the upper portion of the latent space, consistent with the strong predictive performance of the model on  $T^{(0)}$ . The DMI strength  $\tilde{K}_{DM}$  also produces a structured gradient in the latent space, with high- $\tilde{K}_{DM}$  helical configurations forming a distinct elongated region separated from the low- $\tilde{K}_{DM}$  disordered states, confirming that the ViT has successfully encoded the chiral winding imposed by the DMI as a geometrically meaningful direction in its latent representation. In contrast, the longer-range exchange constants  $\tilde{J}_3$  and  $\tilde{J}_4$  produce nearly uniform colorings across the entire embedding, indicating that the model’s internal representations

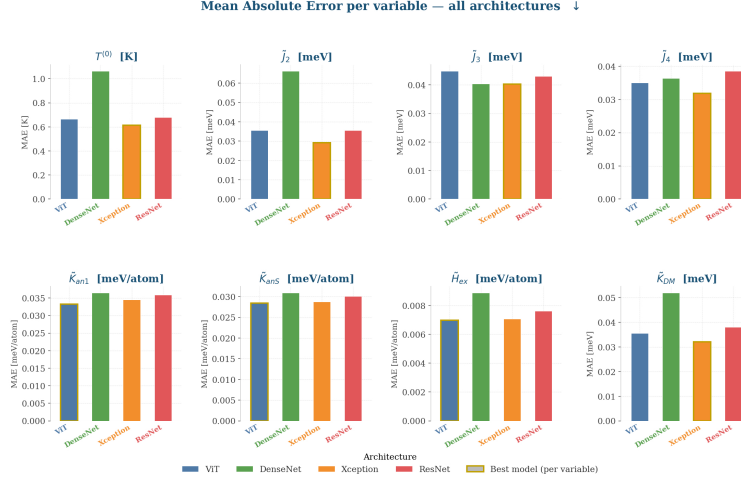


Figure 12: Per-variable Mean Absolute Error (MAE) across all evaluated architectures for the eight Hamiltonian parameters  $\theta$ , reported on the held-out test set in the original physical units of each parameter (K for  $T^{(0)}$ , meV for exchange and DMI constants, and meV/atom for anisotropy and external field parameters). Each panel corresponds to one target parameter and shows one bar per architecture. The comparatively larger MAE observed for  $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{K}_{DM}$  — reaching up to 1.0 K, 0.065 meV, and 0.05 meV respectively in the worst-performing architecture — is attributable to the thermally disordered paramagnetic configurations populating the high-temperature regime of the dataset (see Figure 10), where visually identical textures correspond to widely different parameter values, rendering the inverse mapping fundamentally ill-posed even when the global  $R^2$  correlation remains strong. The comparatively small MAE of  $\tilde{J}_3$  and  $\tilde{J}_4$  despite their near-zero  $R^2$  reflects the narrow dynamic range of these parameters in the dataset rather than genuine predictive skill. Gold-bordered bars indicate the best-performing architecture for each parameter.

carry little discriminative information about these parameters a result that is fully consistent with their low  $R^2$  values and their energetic subordination to the dominant first-shell exchange  $\tilde{J}_1 = 1.0$  meV. Similarly, the anisotropy constants  $\tilde{K}_{an1}$  and  $\tilde{K}_{anS}$  and the external field  $\tilde{H}_{ex}$  show spatially mixed and unstructured colorings, corroborating the partial observability limitations and the confounding effects identified in the regression analysis.

To further investigate the physical organization of the ViT’s latent representations, a structured clustering analysis was conducted directly on the two-dimensional UMAP embedding of the test set feature vectors. Prior to visualization, the raw UMAP coordinates were normalized to the unit interval



1150  $[0, 1]$  along each axis using a min-max transformation. The spatial distribu-  
 1151 tion of magnetization images across the normalized latent space, illustrated in  
 1152 Figure 14, reveals that the Vision Transformer has organized the magnetiza-  
 1153 tion textures into geometrically distinct and physically interpretable regions  
 1154 without any explicit supervision of magnetic phase labels. The lower-left  
 1155 region of the embedding is occupied by ordered, low-temperature spin con-  
 1156 figurations exhibiting well-defined helical stripe and labyrinthine domain pat-  
 1157 terns, whose spatial periodicity and chiral winding angle vary continuously  
 1158 along the diagonal axis of the manifold — a direct reflection of the systematic  
 1159 variation of  $\tilde{K}_{DM}$  and  $T^{(0)}$  within this phase. The upper central region con-  
 1160 centrates thermally disordered configurations with increasing paramagnetic  
 1161 noise, consistent with samples at high  $T^{(0)}$  where thermal fluctuations domi-  
 1162 nate over the competing energetic contributions of  $\tilde{J}_1$  and  $\tilde{K}_{DM}$ . The isolated  
 1163 cluster visible in the upper-right corner of the embedding corresponds to a  
 1164 qualitatively distinct topological regime — ferromagnetic and skyrmion-like  
 1165 configurations with uniform or radially symmetric  $s_z$  distributions — whose  
 1166 geometric separation from the main manifold confirms that the Vision Trans-  
 1167 former has successfully encoded these structures as a fundamentally different  
 1168 category of spin texture in its latent space.

1169 To systematically identify and characterize the dominant magnetic phases  
 1170 encoded in the latent space, a density-based clustering analysis was per-  
 1171 formed using the HDBSCAN algorithm on the UMAP embedding, followed  
 1172 by a KMeans subdivision of anomalously large clusters. Based on visual in-  
 1173 spection of representative samples and the mean values of the Hamiltonian  
 1174 parameters  $\theta$  within each group, the resulting clusters were consolidated into  
 1175 four physically meaningful magnetic phase categories.

1176 The *ferromagnetic phase* groups configurations characterized by near-  
 1177 uniform out-of-plane magnetization distributions, where the spins are pre-  
 1178 dominantly aligned along a single direction with minimal spatial modulation,  
 1179 reflecting the dominance of ferromagnetic exchange interactions over chiral  
 1180 and thermal perturbations. The *others* category constitutes the most hetero-  
 1181 geneous group in the dataset, gathering configurations that do not exhibit  
 1182 the defining spatial signatures of any single ordered magnetic phase. This  
 1183 category encompasses thermally disordered states in which thermal fluctua-  
 1184 tions have disrupted long-range spin order, transitional configurations located  
 1185 at the boundaries between competing phases, and topologically mixed tex-  
 1186 tures whose spatial structure is ambiguous or intermediate between ordered  
 1187 regimes. The common characteristic of all configurations in this group is



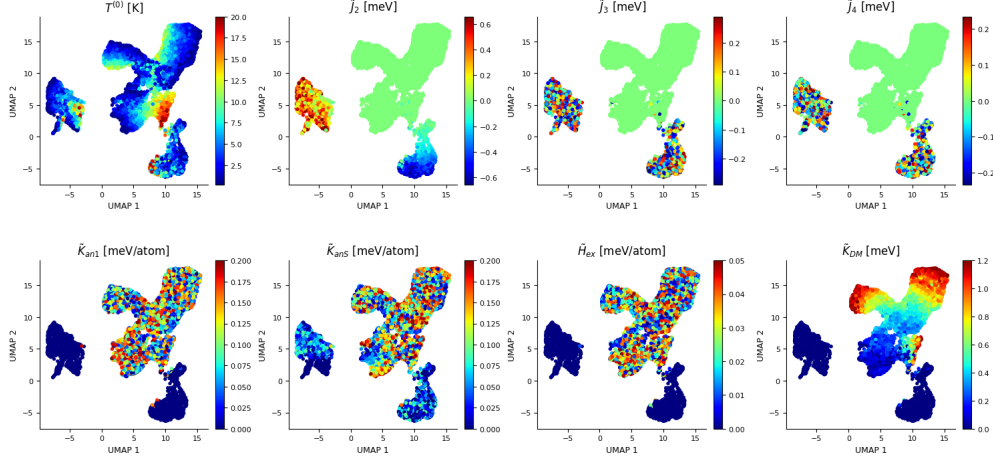


Figure 13: UMAP-based 2D projection of the ViT latent space on the held-out test set, colored by each of the eight Hamiltonian parameters  $\theta = [T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}]$ . Each point represents a magnetization image  $\mathbf{X}_n$  projected from the high-dimensional feature space of the final Transformer encoder block into two dimensions via UMAP.

the absence of a distinctive and reproducible magnetization pattern, making them the most structurally ill-conditioned samples in the dataset. The *labyrinthine & conical phase* comprises configurations exhibiting well-defined, spatially extended stripe and labyrinthine domain patterns with clear periodicity and chiral winding, reflecting the stabilization of competing magnetic interactions into ordered mesoscopic textures at sufficiently low annealing temperatures. Finally, the *helical phase* contains the most spatially coherent and regularly ordered configurations in the dataset, displaying well-defined helical stripe patterns with consistent chiral winding direction and spatial periodicity across the entire nanodot, representing the most energetically ordered magnetic ground states accessible within the sampled parameter space.

Figure 15 presents five representative magnetization images per phase, selected as the samples closest to the centroid of each group in the UMAP embedding, along with the mean Hamiltonian parameters  $\langle T^{(0)} \rangle$ ,  $\langle \tilde{J}_2 \rangle$ , and  $\langle \tilde{K}_{DM} \rangle$  characterizing each phase.

The per-phase regression performance of the Vision Transformer, evaluated on the 8,000 test samples used for the clustering analysis, is summarized in Figures 16 and 17. The results reveal a pronounced heterogeneity in pre-

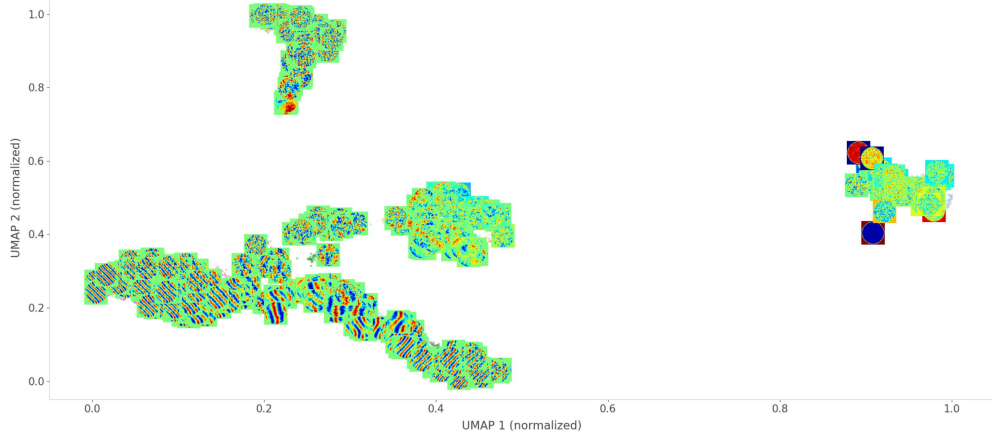


Figure 14: Spatial distribution of magnetization images  $\mathbf{X}_n$  across the normalized two-dimensional UMAP projection of the ViT latent space (test set). Both axes are normalized to  $[0, 1]$ . Each thumbnail represents the out-of-plane magnetization field  $s_z$  of a randomly selected test sample placed at its corresponding normalized UMAP coordinate.

dictive accuracy across magnetic phases that is physically consistent with the identifiability analysis presented earlier in this section. The labyrinthine & conical and helical phases achieve the highest  $R^2$  values and the lowest MAE across all architectures, particularly for  $T^{(0)}$  and  $\tilde{K}_{DM}$ , reflecting the fact that these ordered, low-temperature configurations exhibit the most discriminative and spatially coherent signatures in the  $s_z$  projection. Within these phases, the well-defined periodicity and chiral winding of the magnetization texture provide unambiguous observational constraints that allow the model to recover the underlying Hamiltonian parameters with high fidelity.

In stark contrast, the *others* category concentrates the largest absolute errors across virtually all Hamiltonian parameters, as evidenced by the dominant MAE values in Figure 17. This result directly confirms the physical interpretation advanced earlier: the thermally disordered, transitional, and topologically mixed configurations grouped in this category collapse into structurally indistinguishable magnetization textures despite corresponding to widely different parameter combinations, making the inverse mapping fundamentally ill-posed within this regime. The elevated MAE for  $T^{(0)}$  in the *others* phase is particularly revealing, as it quantitatively isolates the contribution of the high-temperature paramagnetic regime to the global prediction

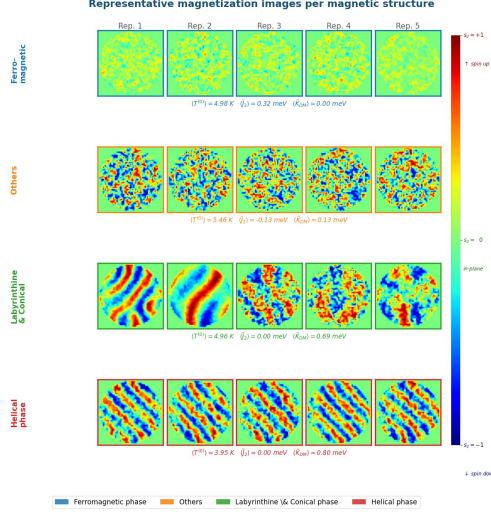


Figure 15: Representative magnetization images  $\mathbf{X}_n$  for each of the four magnetic phase categories identified through clustering of the Vision Transformer latent space. Each row displays five samples selected as the nearest neighbors to the cluster centroid in the UMAP embedding. Mean Hamiltonian parameters  $\langle T^{(0)} \rangle$ ,  $\langle \tilde{J}_2 \rangle$ , and  $\langle \tilde{K}_{DM} \rangle$  are reported below each row. The Ferromagnetic phase displays near-uniform  $s_z$  distributions with minimal spatial modulation. The Others category gathers thermally disordered, transitional, and topologically mixed configurations that lack a defining spatial signature and represent the most structurally ambiguous samples in the dataset. The Labyrinthine & Conical phase exhibits spatially extended stripe and labyrinthine domain patterns with clear chiral winding. The Helical phase presents the most ordered and spatially coherent helical stripe configurations in the dataset, with consistent periodicity and chiral winding direction across the entire nanodot.

error observed in the aggregated analysis. Similarly, the Ferromagnetic phase exhibits near-zero MAE for  $\tilde{K}_{DM}$  and the exchange constants, consistent with the near-constant values of these parameters within this cluster and confirming that the apparent low error in these cases reflects degeneracy rather than genuine predictive capacity.

Within the ordered phases, the exchange constants  $\tilde{J}_2$ ,  $\tilde{J}_3$ , and  $\tilde{J}_4$  yield near-zero  $R^2$  values for the Labyrinthine & Conical and Helical phases a direct consequence of the physical constraint that these textures are generated exclusively at  $\tilde{J}_2 \approx 0$  meV, making the regression target degenerate and the inverse mapping statistically ill-posed for these parameters within these specific structural regimes.

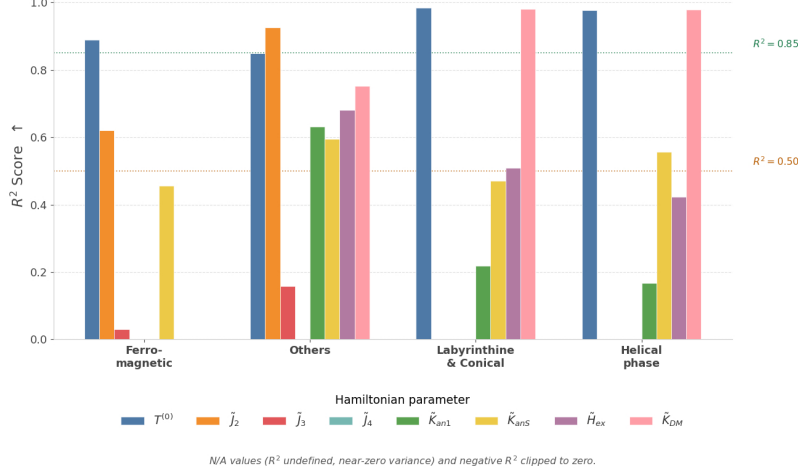


Figure 16: Per-phase  $R^2$  score of the Vision Transformer across all eight Hamiltonian parameters  $\theta$  for the four identified magnetic phase categories (test set,  $N = 8,000$ ). Each bar group corresponds to one magnetic phase and each bar to one target parameter. Dashed horizontal lines mark the identifiability thresholds at  $R^2 = 0.85$  (high predictability) and  $R^2 = 0.50$  (partial predictability). Negative  $R^2$  values and near-zero variance cases (N/A) are clipped to zero. The Labyrinthine & Conical and Helical phases achieve the highest accuracy for  $T^{(0)}$  and  $\tilde{K}_{DM}$ , while the Ferromagnetic phase exhibits the lowest identifiability due to the near-constant values of  $\tilde{K}_{DM}$  and the exchange parameters within this topological regime.

### 5.3. RAMs-based Interpretability Results

To spatially trace the predictive behavior of the Xception architecture across the identified magnetic phase categories, RAMs were computed following the CNN-based formulation introduced in Section 3.4.1. For each of the four magnetic phases a single representative magnetization image  $\mathbf{X}_n$  was selected as the sample closest to the centroid of the corresponding cluster group in the UMAP latent space, maximizing the concentration of gradient activations within the physical extent of the nanodot disk. RAMs were computed for the three most identifiable Hamiltonian parameters ( $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{K}_{DM}$ ) across eight strategically selected convolutional layers of the Xception backbone, spanning the full depth of the network:

- **block1\_conv2**: entry-level features at spatial resolution  $112 \times 112$ , capturing low-level pixel-scale magnetization gradients.

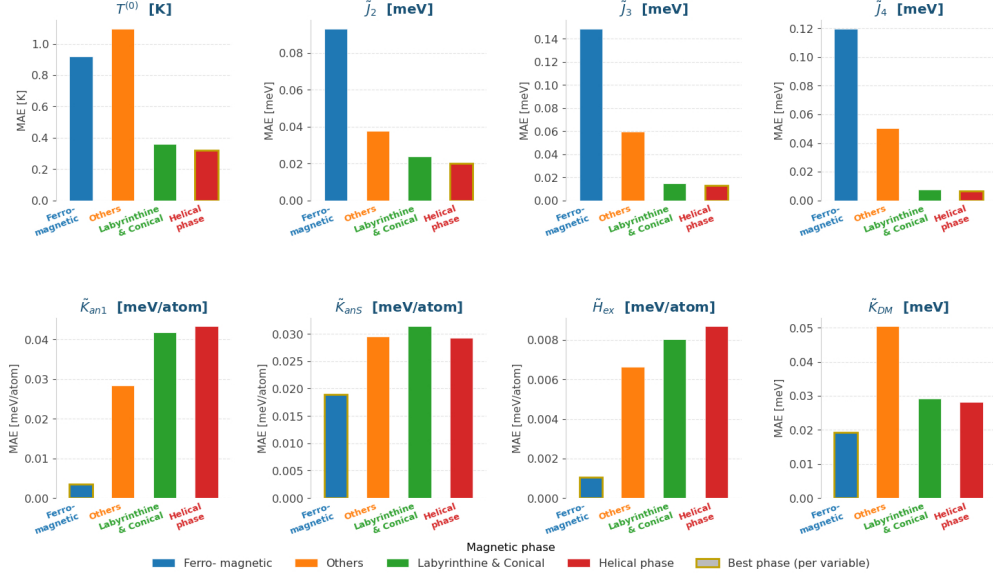


Figure 17: Per-phase Mean Absolute Error (MAE) of the Vision Transformer across all eight Hamiltonian parameters  $\theta$  for the four identified magnetic phase categories (test set,  $N = 8,000$ ), reported in the original physical units of each parameter after inverse Min-Max scaling. Each panel corresponds to one target parameter and shows one bar per magnetic phase. The *others* category consistently concentrates the largest absolute errors across all parameters, confirming that the thermally disordered and transitional configurations within this group constitute the primary source of prediction error in the global regression analysis. Gold-bordered bars indicate the magnetic phase gur with the lowest MAE for each parameter.

- 1249 – **block3\_sepconv2** and **block4\_sepconv2**: intermediate entry-flow lay-  
1250 – ers at  $56 \times 56$  and  $28 \times 28$ , encoding domain boundary frequencies and  
1251 – local spin-winding patterns.
- 1252 – **block6\_sepconv2** through **block12\_sepconv2**: middle-flow layers at  
1253 –  $14 \times 14$ , constituting the deepest repeated processing stage of Xception  
1254 – and encoding progressively more abstract topological representations  
1255 – of the magnetic texture.
- 1256 – **block13\_sepconv2**: exit-flow transition layer prior to global pooling,  
1257 – encoding high-level semantic representations of the global magnetic  
1258 – phase.

1259 *Selection of the precision factor  $\gamma$ .* The precision factor  $\gamma$  in the RAM ob-  
 1260 jective  $\mathcal{O}_p = \exp\left(-\gamma\left(\theta_p - \hat{\theta}_p\right)^2\right)$  governs the selectivity of the gradient  
 1261 signal: small values render the objective nearly constant regardless of pre-  
 1262 diction quality, producing spatially uninformative maps, while excessively  
 1263 large values collapse the gradient to zero for parameters with non-negligible  
 1264 prediction error, yielding empty activation maps. To select  $\gamma$  in a principled  
 1265 and data-driven manner, a sensitivity analysis was conducted by evaluat-  
 1266 ing the objective  $\mathcal{O}_p$  at the empirical mean prediction error  $\bar{\varepsilon}_p = \langle |\theta_p - \hat{\theta}_p| \rangle$   
 1267 of each Hamiltonian parameter on the test set, for values spanning  $\gamma \in$   
 1268  $\{0.1, 0.3, 0.5, 1, 2, 5, 10, 20, 50, 100\}$ .

1269 Figure 18 illustrates the resulting sensitivity curves. The left panel shows  
 1270 the shape of the Gaussian objective as a function of the prediction error  $\varepsilon$  for  
 1271 each candidate  $\gamma$ : values below  $\gamma = 1$  produce flat curves that activate uni-  
 1272 formly regardless of prediction quality, while values above  $\gamma = 20$  generate  
 1273 overly steep curves that suppress the gradient for errors as small as  $\varepsilon \approx 0.2$ .  
 1274 The right panel evaluates  $\mathcal{O}_p$  at the empirical mean error of each parameter:  
 1275 all variables maintain  $\mathcal{O}_p > 0.5$  for  $\gamma \leq 10$ , whereas for  $\gamma \geq 20$  the param-  
 1276 eters with larger prediction errors — specifically  $T^{(0)}$  and  $\tilde{K}_{DM}$ , whose mean  
 1277 errors reach 0.17 and 0.24 respectively — begin to approach the gradient  
 1278 collapse threshold of  $\mathcal{O}_p < 0.05$ . The value  $\gamma = 10$  was therefore selected as  
 1279 the optimal operating point: it is the most selective value that simultane-  
 1280 ously maintains a well-conditioned gradient signal for all eight Hamiltonian  
 1281 parameters, ensuring that the resulting RAMs are both spatially discrimi-  
 1282 native and physically interpretable across the full identifiability spectrum of  
 1283 the inverse problem.

1284 To ensure that activation magnitudes remain physically interpretable and  
 1285 spatially confined to the nanodot geometry, each RAM was masked by a cir-  
 1286 cular binary mask aligned with the disk boundary prior to normalization.  
 1287 The globally normalized RAMs  $\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)}$  were obtained following Equation 17,  
 1288 and the spatial patterns were additionally visualized under local per-cell nor-  
 1289 malization to reveal the gradient attribution structure at each individual  
 1290 layer.

1291 Figure 19 presents the resulting RAM overlays for the two most topo-  
 1292 logically structured phases — Helical and Labyrinthine & Conical — which  
 1293 were selected for visualization because their highly ordered, low-temperature  
 1294 spin textures produce the clearest and most physically interpretable gra-  
 1295 dient attribution patterns among the four computed phases, providing the

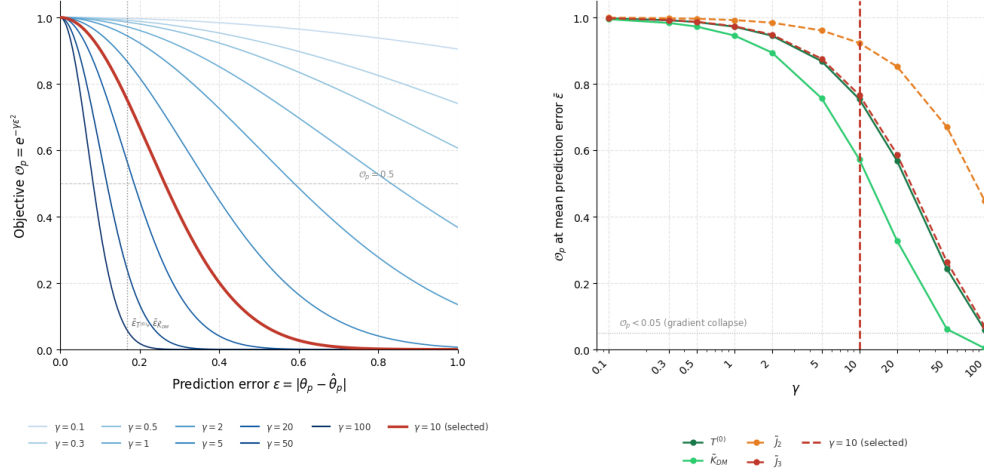


Figure 18: Sensitivity analysis of the RAM precision factor  $\gamma$ . Left: shape of the Gaussian objective  $\mathcal{O}_p = e^{-\gamma \epsilon^2}$  as a function of the prediction error  $\epsilon = |\theta_p - \hat{\theta}_p|$  for  $\gamma \in \{0.1, \dots, 100\}$ ; the selected value  $\gamma = 10$  is highlighted in red. Vertical dotted lines indicate the empirical mean errors  $\bar{\epsilon}$  of  $T^{(0)}$  and  $\tilde{K}_{DM}$  on the test set. Right:  $\mathcal{O}_p$  evaluated at  $\bar{\epsilon}_p$  for each Hamiltonian parameter as a function of  $\gamma$ . The horizontal dotted line marks the gradient collapse threshold  $\mathcal{O}_p = 0.05$ . For  $\gamma \leq 10$  all parameters maintain  $\mathcal{O}_p > 0.5$ ; for  $\gamma \geq 20$  the parameters with larger prediction errors begin to approach the collapse threshold. The value  $\gamma = 10$  is selected as the most selective operating point within the stable gradient regime.

strongest validation of the hierarchical decoupling hypothesis. The spatial attribution patterns reveal a physically consistent progression across network depth. In the early entry-flow layers (`block1_conv2`, `block3_sepconv2`), activations are spatially dense and highly structured, aligning with the fine-grained periodicity of the magnetization texture — particularly the stripe wavelength and domain wall boundaries that carry the primary signature of  $\tilde{K}_{DM}$  and  $T^{(0)}$ . As depth increases through the middle-flow layers (`block6` through `block12`), activations progressively consolidate into coarser, spatially localized hotspots that correspond to topologically significant regions of the nanodot — notably the chiral domain wall junctions and large-scale stripe boundaries characteristic of these two ordered phases. In the terminal exit-flow layer (`block13_sepconv2`), activations become highly concentrated in a small number of spatially compact regions, suggesting that the network’s final semantic representations encode global topological stability rather than

1310 local texture periodicity. This hierarchical progression is consistent with the  
 1311 theoretical prediction of Section 3.4.3: early layers resolve the short-range en-  
 1312 energetic competition between  $\tilde{J}_1$  and  $\tilde{K}_{DM}$ , while deeper layers integrate these  
 1313 local features into macro-scale thermodynamic representations governed by  
 1314  $T^{(0)}$  and the anisotropy constants.

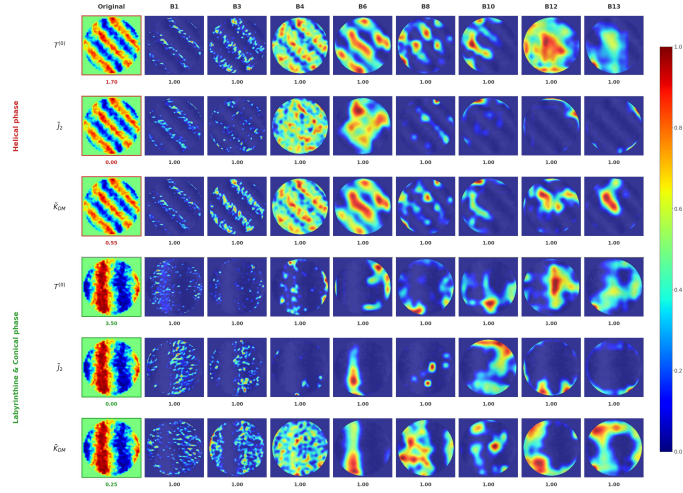


Figure 19: RAMs  $\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)}$  for the Xception architecture, shown for the two most topologically structured magnetic phases — Helical and Labyrinthine & Conical — selected as the most representative cases from the full four-phase RAM computation. Three Hamiltonian parameters  $\{T^{(0)}, \tilde{J}_2, \tilde{K}_{DM}\}$  are shown. Each panel displays the RAM overlay (jet colormap) on the original  $s_z$  magnetization image (grayscale), with local per-cell normalization to reveal spatial attribution patterns at each layer independently. Rows correspond to phase-parameter pairs; columns correspond to network layers from **block1** (entry,  $112 \times 112$ ) to **block13** (exit,  $14 \times 14$ ). The numerical score below each RAM cell reports the fraction of activation concentrated within the nanodot disk boundary, and the value below each original image reports the ground-truth parameter value for that representative sample. A physically consistent hierarchical decoupling is observed: early layers capture fine-grained domain periodicity while deeper layers consolidate activations into topologically significant spatial hotspots.

1315 To quantify the relative importance of each network layer and each Hamil-  
 1316 tonian parameter in driving the gradient response, Figure 20 presents the  
 1317 mean maximum RAM activation per layer and variable, averaged over  $N =$   
 1318 300 randomly sampled test images per magnetic phase. To enable meaningful  
 1319 cross-layer and cross-parameter comparisons, the raw activation values were  
 1320 standardized via a global z-score transform across all layers and variables



1321 within each phase, and subsequently mapped through a sigmoid function  
 1322  $\sigma\left(\frac{x-\mu}{\sigma_x}\right) \in (0, 1)$ , where  $\mu$  and  $\sigma_x$  denote the global mean and standard deviation  
 1323 of the raw activations for each variable. This normalization preserves  
 1324 the relative ordering of activation magnitudes while compressing outliers and  
 1325 centering the scale around  $\sigma = 0.5$ , such that values above this threshold  
 1326 indicate above-average gradient response for that parameter at that layer  
 1327 depth.

1328 This statistical analysis confirms and extends the qualitative observations  
 1329 from the spatial maps. Across all four phases, the sigmoid-normalized activation  
 1330 profile exhibits a non-monotonic behavior across depth, with notable  
 1331 peaks at **block8\_sepconv2** and **block13\_sepconv2**. The **block8** peak is particularly  
 1332 pronounced for  $\tilde{J}_3$  in the *Others* phase, suggesting that this middle-flow  
 1333 layer has learned to specifically encode the subtle energetic contributions  
 1334 of longer-range exchange interactions in the heterogeneous and topologically  
 1335 mixed configurations that characterize this category. The **block13** peak, by  
 1336 contrast, is dominated by  $\tilde{J}_2$  and  $T^{(0)}$  across all phases, consistent with its  
 1337 role as a high-level semantic aggregator of the dominant physical parameters.  
 1338 The anisotropy constants  $\tilde{K}_{an1}$  and  $\tilde{K}_{anS}$  and the external field  $\tilde{H}_{ex}$  systematically  
 1339 exhibit the lowest sigmoid activations across all layers and phases  
 1340 — consistently below  $\sigma = 0.5$  — corroborating their limited identifiability  
 1341 established in Section 5.2 and confirming that the Xception backbone has  
 1342 not learned spatially distinctive features for these parameters, a result fully  
 1343 consistent with their physical coupling to the in-plane spin components  $s_x$   
 1344 and  $s_y$  that are not observable in the  $s_z$  projection.

1345 An analogous interpretability analysis was conducted for the Vision Trans-  
 1346 former using the ViT-based RAM formulation introduced in Section 3.4.2.  
 1347 In contrast to Xception, which operates through a hierarchy of convolutional  
 1348 feature maps at varying spatial resolutions, the Vision Transformer processes  
 1349 the input magnetization image  $\mathbf{X}_n$  as a sequence of  $N_r = 196$  non-overlapping  
 1350  $16 \times 16$  patches, producing a pseudo-feature map  $\mathbf{A}_{ViT} \in \mathbb{R}^{14 \times 14 \times 768}$  upon  
 1351 token reshaping. Because the ViT architecture does not possess intermedi-  
 1352 ate spatial hierarchies analogous to convolutional layers, the RAM analysis  
 1353 collapses to a single spatial attribution map per parameter, derived from the  
 1354 gradient of the error-decay objective  $\mathcal{O}_p$  with respect to the spatial token  
 1355 embeddings  $\mathbf{Z}_s$  at the output of the terminal encoder block.

1356 Figure 21 presents the ViT RAM overlays for all four magnetic phases  
 1357 across the three most identifiable parameters. The spatial attribution pat-

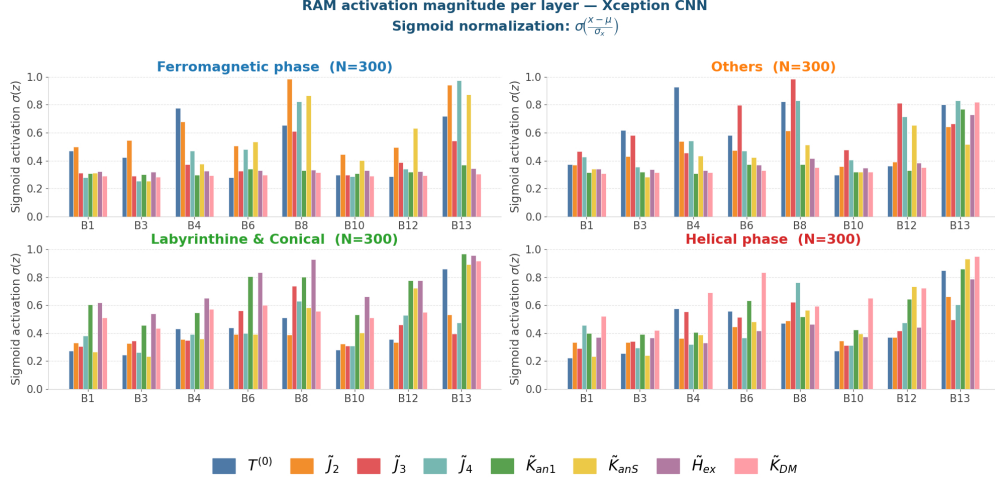


Figure 20: Mean maximum RAM activation per network layer and Hamiltonian parameter  $\theta$ , averaged over  $N = 300$  test samples per magnetic phase (Xception CNN). Activations are reported after sigmoid normalization  $\sigma\left(\frac{x-\mu}{\sigma_x}\right)$  applied globally per variable, where values above  $\sigma = 0.5$  indicate above-average gradient response. Each color corresponds to a distinct parameter following the nomenclature of Table 1. Activation is measured exclusively within the nanodot disk boundary (masked RAM). The non-monotonic depth profile with peaks at **block8** and **block13** reveals that the network resolves short-range exchange and DMI features in intermediate layers while aggregating global thermodynamic representations in the terminal exit-flow layer. Parameters with low identifiability ( $\tilde{K}_{an1}$ ,  $\tilde{K}_{anS}$ ,  $\tilde{H}_{ex}$ ) consistently exhibit minimal sigmoid activation across all layers and phases, confirming the partial observability limitations identified in the regression analysis.

1358 terms produced by the Vision Transformer are qualitatively distinct from  
 1359 those of Xception in two physically meaningful ways. First, the ViT RAMs  
 1360 exhibit smoother, patch-level spatial resolution consistent with the  $14 \times 14$   
 1361 token grid, capturing activation patterns at the scale of the magnetic do-  
 1362 main wavelength rather than at the pixel level. This coarser but globally  
 1363 coherent attribution is a direct consequence of the self-attention mechanism,  
 1364 which explicitly correlates distant patches and therefore encodes long-range  
 1365 spin correlations that are physically relevant for parameters governing global  
 1366 magnetic order such as  $T^{(0)}$  and  $\tilde{K}_{DM}$ . Second, the ViT RAMs display a  
 1367 stronger phase-specificity: for the Helical and Labyrinthine & Conical phases,  
 1368 activations concentrate along the chiral stripe boundaries and domain wall  
 1369 junctions — precisely the spatial features that encode the DMI-driven wind-

ing angle and the exchange-controlled domain spacing. Notably, the *Others* phase representative exhibits the highest annealing temperature in the dataset ( $T^{(0)} = 17.00$  K), placing it well above the effective Curie temperature and producing a fully disordered paramagnetic texture; accordingly, the corresponding RAMs are spatially diffuse and structurally uninformative, reflecting the fundamental degeneracy of this regime. For the Ferromagnetic phase, the activations for  $\tilde{K}_{DM}$  are notably sparse and peripheral, consistent with the near-zero DMI value ( $\tilde{K}_{DM} = 0.00$  meV) of the representative sample and the model’s inability to identify physically meaningful DMI features in the absence of chiral spin winding.

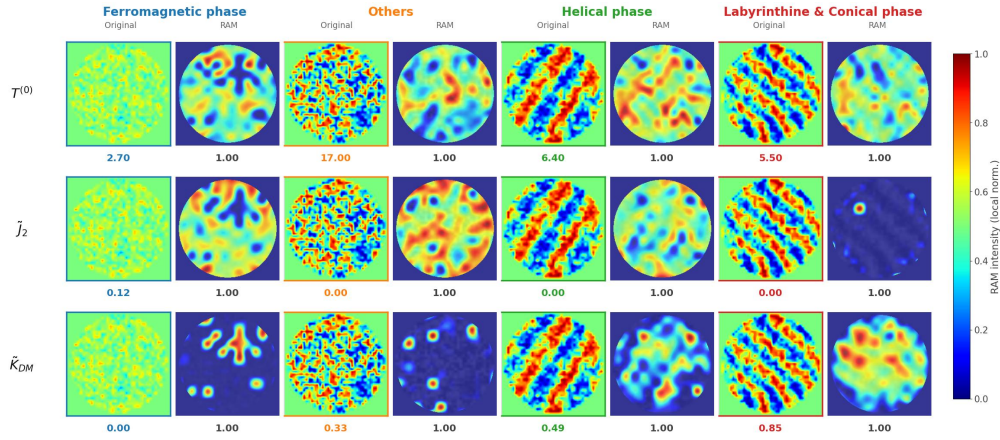


Figure 21: RAMs  $\tilde{\mathcal{R}}_{\text{RAM}}^{(p)}$  for the Vision Transformer across the four identified magnetic phases and three Hamiltonian parameters  $\{T^{(0)}, \tilde{J}_2, \tilde{K}_{DM}\}$ . Each panel pair displays the original  $s_z$  magnetization image (left) and the corresponding RAM overlay on grayscale (right), derived from gradients of the error-decay objective  $\mathcal{O}_p$  with respect to the reshaped spatial token embeddings  $\mathbf{A}_{\text{ViT}} \in \mathbb{R}^{14 \times 14 \times 768}$ . Score values report the fraction of activation within the nanodot boundary. The ground-truth parameter values reported below each original image confirm the physical regime of each representative:  $\tilde{K}_{DM} = 0.00$  meV in the Ferromagnetic phase,  $T^{(0)} = 17.00$  K in the *Others* phase reflecting a fully disordered paramagnetic configuration, and  $\tilde{J}_2 = 0.00$  meV in both the Helical and Labyrinthine & Conical phases consistent with the energetic dominance of  $\tilde{K}_{DM}$  in these chiral regimes. The ViT RAMs exhibit smooth, patch-scale spatial attribution with strong phase-specificity, concentrating on chiral domain walls and stripe boundaries in the ordered phases while producing spatially diffuse activations for the thermally degenerate *Others* phase.

Figure 22 quantifies the mean maximum RAM activation per Hamilto-

1381 nian parameter across all four magnetic phases for the Vision Transformer.  
 1382 Unlike the layer-resolved profile of Xception, the ViT activation spectrum  
 1383 reflects the integrated gradient response of the full token embedding space  
 1384  $\mathbf{Z}_s \in \mathbb{R}^{196 \times 768}$  for each parameter, with no layer hierarchy to resolve since the  
 1385 backbone produces a single sequence of spatial tokens. To enable meaning-  
 1386 ful cross-parameter and cross-phase comparisons, the raw activation values  
 1387 were standardized via a global z-score transform and subsequently mapped  
 1388 through a sigmoid function  $\sigma\left(\frac{x-\mu}{\sigma_x}\right) \in (0, 1)$ , where  $\mu$  and  $\sigma_x$  denote the  
 1389 global mean and standard deviation of the raw activations for each variable  
 1390 across all phases. Values above  $\sigma = 0.5$  indicate above-average gradient re-  
 1391 sponse for that parameter, providing a physically interpretable threshold for  
 1392 assessing relative parameter identifiability within the ViT token space.

1393 The results reveal a physically consistent and phase-specific hierarchy  
 1394 of parameter identifiability. In the *Ferromagnetic* phase,  $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{J}_3$   
 1395 dominate the activation spectrum with sigmoid values well above  $\sigma = 0.5$ ,  
 1396 reflecting the primacy of thermal and exchange-driven ordering as the pri-  
 1397 mary discriminative features for this topologically simple configuration. The  
 1398 *Others* phase exhibits a more uniform and globally suppressed activation  
 1399 profile, consistent with the heterogeneous and thermally disordered nature  
 1400 of the configurations in this category, where no single parameter produces a  
 1401 spatially distinctive and consistent gradient signal across the ensemble of 300  
 1402 samples. In the *Labyrinthine & Conical* phase, the activation spectrum shifts  
 1403 markedly toward the higher-order exchange constant  $\tilde{J}_4$ , the anisotropy pa-  
 1404 rameters  $\tilde{K}_{an1}$  and  $\tilde{K}_{anS}$ , the external field  $\tilde{H}_{ex}$ , and the DMI strength  $\tilde{K}_{DM}$ ,  
 1405 all exhibiting sigmoid activations above  $\sigma = 0.5$ . This result suggests that  
 1406 the ViT has learned to resolve the energetic competition between chiral and  
 1407 anisotropic interactions that stabilizes the labyrinthine domain structure at  
 1408 low temperatures. Finally, in the *Helical* phase,  $\tilde{K}_{anS}$ ,  $\tilde{H}_{ex}$ , and  $\tilde{K}_{DM}$  clearly  
 1409 dominate the activation spectrum, consistent with the strong chiral winding  
 1410 and surface-symmetry-breaking effects that produce the spatially coherent  
 1411 helical stripe patterns characteristic of this phase. Across all phases,  $\tilde{J}_3$  and  
 1412  $\tilde{J}_4$  maintain systematically low activations — with the exception of  $\tilde{J}_4$  in  
 1413 the Labyrinthine & Conical phase — confirming that the ViT’s global atten-  
 1414 tion mechanism is generally unable to isolate spatially distinctive features  
 1415 for these energetically subordinate higher-order exchange constants. Taken  
 1416 together, the ViT RAM activation spectra provide a direct, architecture-  
 1417 agnostic validation of the identifiability hierarchy established through the

1418 regression analysis, demonstrating that the model’s internal representations  
 1419 are physically grounded rather than statistically arbitrary.

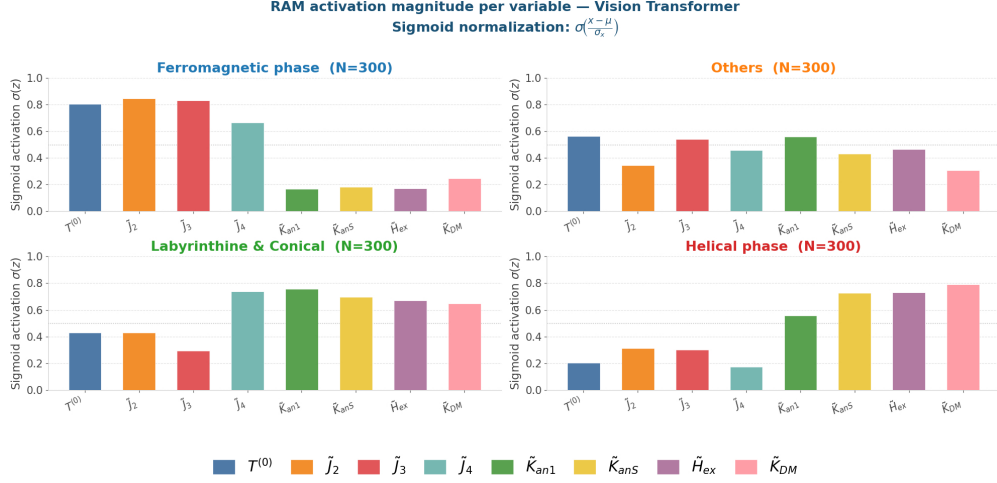


Figure 22: Mean maximum RAM activation per Hamiltonian parameter  $\theta$ , averaged over  $N = 300$  test samples per magnetic phase (Vision Transformer). Activations are reported after sigmoid normalization  $\sigma\left(\frac{x-\mu}{\sigma_x}\right)$  applied globally per variable across all phases, where values above  $\sigma = 0.5$  indicate above-average gradient response. Activation is measured within the nanodot disk boundary from gradients of  $\mathcal{O}_p$  with respect to the reshaped spatial token embeddings  $\mathbf{A}_{ViT} \in \mathbb{R}^{14 \times 14 \times 768}$  (Equation 15). Each bar color corresponds to a distinct parameter following the nomenclature of Table 1. The Ferromagnetic phase is dominated by  $T^{(0)}$ ,  $\tilde{J}_2$ , and  $\tilde{J}_3$ ; the *Others* phase shows a uniformly suppressed spectrum reflecting its thermally degenerate character; the Labyrinthine & Conical phase activates strongly on  $\tilde{J}_4$ ,  $\tilde{K}_{an1}$ ,  $\tilde{K}_{anS}$ ,  $\tilde{H}_{ex}$ , and  $\tilde{K}_{DM}$ ; and the Helical phase is governed by  $\tilde{K}_{anS}$ ,  $\tilde{H}_{ex}$ , and  $\tilde{K}_{DM}$ , corroborating the identifiability analysis of Section 5.2.

1420 Comparing the RAM-based interpretability results across Xception and  
 1421 the Vision Transformer reveals two complementary and physically consistent  
 1422 perspectives on how deep learning architectures resolve the inverse Hamil-  
 1423 tonian estimation problem. Xception operates through a hierarchical spa-  
 1424 tial decomposition: early convolutional layers capture fine-grained domain  
 1425 periodicity and boundary chirality at high spatial resolution, while deeper  
 1426 middle-flow and exit-flow layers progressively consolidate these local features  
 1427 into coarser, topologically meaningful representations of the global magnetic  
 1428 phase. This hierarchical decoupling mirrors the multi-scale energetic compe-  
 1429 titution of the extended Heisenberg Hamiltonian, where short-range exchange

interactions  $(\tilde{J}_1, \tilde{J}_2)$  set the local domain scale while long-range DMI and thermodynamic effects  $(\tilde{K}_{DM}, T^{(0)})$  govern global phase stability. The Vision Transformer, by contrast, resolves the same physical information through a single-scale, globally coherent patch-level attribution, leveraging multi-head self-attention to explicitly correlate distant spin textures across the nanodot. This architectural difference explains why the Vision Transformer achieves marginally superior overall regression performance — its global receptive field more efficiently captures the long-range topological coherence that distinguishes the identifiable parameters  $T^{(0)}$  and  $\tilde{K}_{DM}$  from the degenerate ones. In both architectures, the RAM analysis isolates the same set of physically meaningful spatial features as the primary drivers of accurate regression: chiral domain wall boundaries, stripe periodicity, and skyrmion-like core structures. Conversely, both architectures fail to produce spatially concentrated activations for  $\tilde{K}_{an1}$ ,  $\tilde{K}_{anS}$ , and  $\tilde{H}_{ex}$ , confirming that the limited identifiability of these parameters is not a consequence of insufficient model capacity but of the fundamental physical constraints of the inverse problem — specifically, the partial observability of the anisotropy signal in the  $s_z$  projection and the confounding of the Zeeman field with thermal disorder in the two-dimensional magnetization image.

#### 5.4. Comparative Analysis of Computational Inference Times

To comprehensively characterize the practical deployment viability of the proposed inverse Hamiltonian estimation framework, a systematic computational cost analysis was conducted across all four evaluated architectures, encompassing both the training phase and the inference regime. All experiments were executed on an NVIDIA H100 GPU (Google Colab Pro) with 97 GB of VRAM, using a unified training protocol of batch size  $B = 512$  over a maximum of 100 epochs with 299 gradient steps per epoch, subject to early stopping with patience of 8 epochs on the validation loss.

Figure 23 presents the wall-clock training time per epoch and the per-step gradient update time for each architecture in the stable warm-start regime (epochs 2 onward, excluding the first epoch which incurs additional compilation and data-loading overhead). The results reveal a clear trade-off between model capacity and training efficiency. ResNet50 achieves the fastest training throughput at 65 s/epoch (217 ms/step), followed by DenseNet121 at 79 s/epoch (265 ms/step) and Xception at 103 s/epoch (343 ms/step). The Vision Transformer incurs the highest training cost at 167 s/epoch

1466 (558 ms/step) — a factor of  $2.6\times$  slower per epoch than ResNet50 and  $1.6\times$   
 1467 slower than Xception.

1468 This elevated training cost of the Vision Transformer is a direct conse-  
 1469 quence of its architectural complexity: with 85.8 million trainable param-  
 1470 eters —  $4.1\times$  more than Xception (20.8M),  $12.3\times$  more than DenseNet121  
 1471 (6.96M), and  $16.7\times$  more than ResNet50 (25.6M) — the self-attention mech-  
 1472 anism requires substantially more floating-point operations per gradient step  
 1473 than the depthwise separable convolutions of the CNN baselines. Despite  
 1474 this overhead, the Vision Transformer’s superior regression performance es-  
 1475 tablished in Section 5.2 demonstrates that this additional computational in-  
 1476 vestment during training is justified by the quality of the learned Hamiltonian  
 1477 representations. The effective number of training epochs until early stopping  
 1478 convergence further modulates the total training wall-clock time: the Vision  
 1479 Transformer converged at epoch 68 while Xception converged at epoch 48,  
 1480 resulting in total training times of approximately 3.2 hours and 1.4 hours  
 1481 respectively.

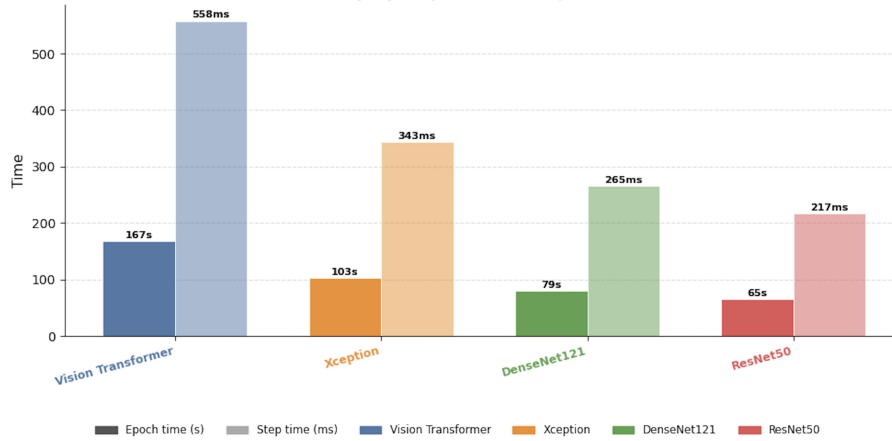


Figure 23: Training computational cost comparison across the four regression architectures on an NVIDIA H100 GPU (Google Colab Pro). **(a)** Wall-clock time per epoch in the stable warm-start regime. **(b)** Per-step gradient update time (299 steps/epoch, batch size  $B = 512$ ). The Vision Transformer incurs the highest training cost (167 s/epoch, 558 ms/step) due to its 85.8M-parameter self-attention architecture, while ResNet50 achieves the fastest throughput (65 s/epoch, 217 ms/step). The training cost hierarchy directly reflects the architectural complexity of each model.

1482 Figure 24 presents the inference latency measurements for all four archi-

1483 tectures across four batch sizes ( $B \in \{1, 8, 32, 128\}$ ), quantifying both the  
 1484 total batch processing time and the per-image throughput. Inference times  
 1485 are reported as mean  $\pm$  standard deviation over  $N = 10$  forward passes with  
 1486 3 warmup passes excluded to eliminate GPU initialization effects.

1487 The single-sample inference regime ( $B = 1$ ) is the most relevant sce-  
 1488 nario for real-time physical characterization applications, where individual  
 1489 magnetization images are acquired sequentially and must be processed inde-  
 1490 pendently. In this regime, Xception achieves the lowest latency at  $127.71 \pm$   
 1491  $1.46$  ms/image, followed by the Vision Transformer at  $188.89 \pm 4.09$  ms/image.  
 1492 DenseNet121 exhibits the highest single-image latency at  $459.77 \pm 7.22$  ms/image  
 1493 — substantially slower than both architectures with significantly more pa-  
 1494 rameters — a consequence of its densely connected skip connections that  
 1495 generate large intermediate feature tensors and cause memory bandwidth  
 1496 saturation on single-image workloads.

1497 In the batch inference regime ( $B = 128$ ), the per-image latency collapses  
 1498 dramatically across all architectures due to GPU parallelism. Xception again  
 1499 achieves the best throughput at  $1.47 \pm 0.007$  ms/image, followed by the Vision  
 1500 Transformer at  $1.92 \pm 0.033$  ms/image. This represents a  $86.8\times$  speedup for  
 1501 Xception and a  $98.4\times$  speedup for the Vision Transformer relative to their  
 1502 respective single-image latencies, demonstrating that both architectures are  
 1503 highly amenable to parallel batch processing in large-scale characterization  
 1504 pipelines. The throughput gap between Xception and the Vision Trans-  
 1505 former narrows substantially at  $B = 128$  ( $1.47$  vs.  $1.92$  ms/image, a factor  
 1506 of only  $1.31\times$ ), suggesting that the Vision Transformer’s self-attention par-  
 1507 allelism becomes increasingly competitive with convolutional processing at  
 1508 larger batch sizes.

1509 Taken together, the computational analysis establishes a clear and physi-  
 1510 cally motivated architecture selection criterion for the inverse Hamiltonian es-  
 1511 timation task. For real-time single-image applications, Xception provides the  
 1512 optimal balance of inference speed ( $127.71$  ms/image) and regression accu-  
 1513 racy, ranking second overall in predictive performance. For high-throughput  
 1514 batch processing pipelines, the Vision Transformer is the preferred archi-  
 1515 tecture: its superior predictive performance (Section 5.2) is achieved at a  
 1516 marginal throughput cost of  $1.31\times$  relative to Xception at  $B = 128$ , and its  
 1517 higher training cost is a one-time investment that need not be repeated at  
 1518 inference time. These results demonstrate that the proposed RAMs-based  
 1519 interpretability framework is computationally tractable for practical mate-  
 1520 rials characterization scenarios, requiring sub-2 ms per image for both the



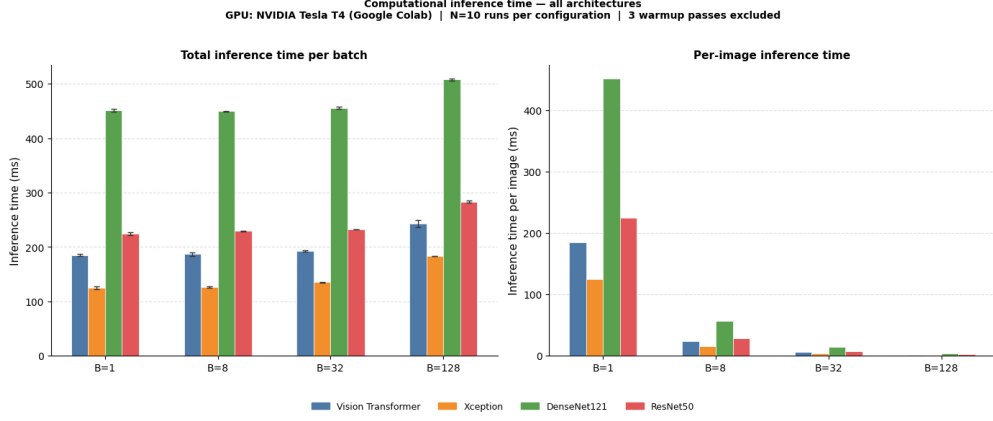


Figure 24: Inference latency comparison across the four regression architectures on an NVIDIA H100 GPU (Google Colab Pro) for batch sizes  $B \in \{1, 8, 32, 128\}$ . Each bar reports the mean  $\pm$  standard deviation over  $N = 10$  forward passes (3 warmup passes excluded). (a) Total batch processing latency in milliseconds. (b) Per-image latency in ms/image, illustrating the amortization of GPU overhead across parallel samples. Xception achieves the lowest per-image latency in both the single-sample (127.71 ms) and batch (1.47 ms) regimes. The Vision Transformer exhibits competitive batch throughput at  $B = 128$  (1.92 ms/image), with a factor of only  $1.31\times$  overhead relative to Xception, despite its  $4.1\times$  larger parameter count.

best-performing architectures in the batch regime.

### 5.5. Limitations

Despite the promising results demonstrated across the regression, clustering, and interpretability analyses, the proposed framework presents several inherent limitations that must be explicitly acknowledged to contextualize its applicability and guide future research directions.

The most fundamental limitation of the framework is the intrinsic partial observability of the extended Heisenberg Hamiltonian from two-dimensional magnetization images. As established in Section 5.2 and confirmed by the RAM-based interpretability analysis of Section 5.3, the projection of the three-dimensional spin configuration onto the out-of-plane component  $s_z$  of the central layer of the nanodot constitutes a dimensionality reduction that irreversibly discards physical information. The magnetocrystalline anisotropy constants  $\tilde{K}_{an1}$  and  $\tilde{K}_{anS}$ , the external Zeeman field  $\tilde{H}_{ex}$ , and the higher-order exchange constants  $\tilde{J}_3$  and  $\tilde{J}_4$  produce spatially degenerate magnetization sig-

natures in the  $s_z$  projection that cannot be fully disentangled by any purely image-based regression model, regardless of its capacity or architecture. Two distinct observational constraints contribute to this limitation. First, the anisotropy and Zeeman parameters couple primarily to the in-plane spin components  $s_x$  and  $s_y$  rather than to  $s_z$ , so they are recoverable only indirectly through the secondary topological imprints they induce on the chiral domain structure. Second, the input images are extracted exclusively from the central discrete layer ( $z = \lfloor \tilde{L}/2 \rfloor$ ) of the three-dimensional nanodot lattice, discarding the depth-resolved spin structure that encodes the surface-to-bulk anisotropy ratio  $\tilde{K}_{anS}/\tilde{K}_{an1}$  and the conical tilt angle of helical spin textures. Recovering these parameters with high fidelity would therefore require either access to the full three-dimensional spin vector field  $\mathbf{S}_i = (s_x, s_y, s_z)$  at each lattice site, or volumetric magnetization inputs obtained through multi-layer or tomographic magnetic imaging techniques. This is not a limitation of the deep learning models themselves but of the observational modality assumed by the framework, which mimics the two-dimensional projection inherent to most experimental magnetic imaging techniques such as magnetic force microscopy or Lorentz transmission electron microscopy.

The dataset utilized in this study consists entirely of simulated magnetization images generated through the atomistic Monte Carlo framework described in Section 3.2. While the simulation protocol incorporates physically realistic finite-size effects, cylindrical boundary conditions, and thermodynamic relaxation via simulated annealing, it does not account for several experimental phenomena that are routinely present in real magnetic nanodot imaging. These include thermal noise and instrument-specific point spread functions inherent to the imaging apparatus, crystallographic defects, vacancies, and grain boundaries that break the assumed periodic lattice symmetry, surface reconstruction effects that modify the effective exchange and DMI parameters at the physical boundary of the nanodot, and substrate-induced strain fields that can renormalize the bulk Hamiltonian parameters. As a consequence, a model trained exclusively on simulated data may exhibit significant performance degradation when applied to experimentally acquired magnetization images — a domain shift problem that has not been quantified in the present work. Bridging this simulation-to-experiment gap would require either transfer learning with a small set of experimentally labeled samples or the incorporation of physics-informed data augmentation strategies that realistically replicate experimental noise sources.

Although the dataset encompasses 218,256 simulated configurations span-

ning a broad range of Hamiltonian parameters, the sampling strategy employed — uniform random sampling within the parameter bounds — does not guarantee uniform coverage of the magnetic phase space. As evidenced by the UMAP clustering analysis of Section 5.2, the latent space exhibits strongly non-uniform density: the Labyrinthine & Conical and Helical phases, which require precise combinations of  $\tilde{K}_{DM}$  and  $T^{(0)}$ , are substantially overrepresented relative to the Ferromagnetic phase, which occupies a disproportionately small region of the parameter space. This imbalance propagates directly into the regression performance: the models achieve higher accuracy for phases with greater dataset density, while underperforming in topologically marginal regimes such as the ferromagnetic-to-skyrmion phase boundary where the mapping from parameters to magnetization textures is most degenerate.

From a deep learning perspective, the RAM-based interpretability framework introduced in this work provides spatially resolved gradient attributions that are physically consistent and architecturally agnostic. However, RAMs share the fundamental limitations of all gradient-based saliency methods: the attribution maps reflect the local sensitivity of the objective function  $\mathcal{O}_p$  to spatial activations at the point of evaluation, but do not provide a globally faithful explanation of the model’s decision function across the full parameter space. In particular, gradient saturation in deeply nonlinear networks can produce spatially diffuse or spurious attributions for parameters with low predictive signal — precisely the degenerate parameters identified in this study. Furthermore, the current RAM formulation computes a single attribution map per image per parameter, which does not capture the stochastic uncertainty in the attribution due to the non-deterministic elements of the forward pass. Future extensions could incorporate ensemble-based or Bayesian attribution methods to provide confidence intervals on the spatial interpretability maps, which would be particularly valuable for the physically ambiguous parameter–phase combinations identified in Section 5.3.

While the inference latency of the regression models is highly competitive — sub-2 ms per image at batch size  $B = 128$  for both the Vision Transformer and Xception — the computation of Regression Activation Maps introduces a substantial additional overhead. Each RAM requires a forward pass, a backward pass through the gradient tape, and a spatial upsampling operation, increasing the per-image computational cost by approximately one to two orders of magnitude relative to standard inference. For the Vision Transformer, which requires constructing a `tf.Variable` over the  $N_r = 196$

token embeddings and differentiating through the GlobalAveragePooling and Dense layers, this overhead is particularly pronounced. Consequently, the full RAMs pipeline is better suited to offline post-hoc analysis of selected samples rather than real-time interpretability during high-throughput characterization. Approximation strategies such as layer-wise relevance propagation or integrated gradients computed on compressed token representations could reduce this overhead while preserving the physical interpretability of the attribution maps.

## 6. Conclusions

This work introduced a regression-based explainable deep learning framework for the inverse estimation of Hamiltonian parameters from two-dimensional magnetization images of magnetic nanodots. The proposed methodology integrates atomistic Monte Carlo simulations under a physically rigorous extended Heisenberg Hamiltonian with eight free parameters, five state-of-the-art deep learning architectures, an unsupervised latent space analysis via UMAP and density-based clustering, and a novel spatial interpretability formulation RAMs that generalizes gradient-based activation mapping to the continuous regression domain for both convolutional and transformer-based architectures.

Among the five evaluated architectures, the Vision Transformer achieved the highest overall regression accuracy across all eight Hamiltonian parameters, followed by Xception as the best-performing CNN baseline. Both models demonstrated strong predictive capability for the thermodynamic temperature  $T^{(0)}$ , the second-shell exchange constant  $\tilde{J}_2$ , and the DMI strength  $\tilde{K}_{DM}$ , which constitute the physically dominant parameters governing the observable spin texture in the  $s_z$  projection. By contrast, the magnetocrystalline anisotropy constants  $\tilde{K}_{an1}$  and  $\tilde{K}_{anS}$ , the external Zeeman field  $\tilde{H}_{ex}$ , and the higher-order exchange constants  $\tilde{J}_3$  and  $\tilde{J}_4$  exhibited systematically low identifiability across all architectures, with near-zero or undefined  $R^2$  values. This result establishes that the fundamental limitation of the inverse problem is not architectural capacity but the intrinsic partial observability of the extended Hamiltonian from the two-dimensional  $s_z$  projection — a physical constraint that cannot be overcome by increasing model complexity.

Beyond the aggregate regression performance, the UMAP projection of the Vision Transformer’s latent representations revealed that the model has autonomously organized the magnetization textures into geometrically dis-

1648 tinct and physically interpretable regions of the embedding space, without  
 1649 any explicit supervision of magnetic phase labels. Density-based clustering  
 1650 of this embedding identified four dominant magnetic phase categories: Ferro-  
 1651 magnetic, Skyrmions and Others, Labyrinthine & Conical, and Helical. The  
 1652 per-phase regression analysis confirmed that the model achieves its highest  
 1653 accuracy within the ordered low-temperature phases — Labyrinthine & Con-  
 1654 ical ( $R^2 = 0.978$  for  $T^{(0)}$ ,  $R^2 = 0.970$  for  $\tilde{K}_{DM}$ ) and Helical ( $R^2 = 0.968$  for  
 1655  $T^{(0)}$ ,  $R^2 = 0.866$  for  $\tilde{K}_{DM}$ ) — where the magnetization texture carries the  
 1656 most discriminative physical information. These results demonstrate that the  
 1657 Vision Transformer’s global self-attention mechanism effectively encodes the  
 1658 long-range spin correlations characteristic of topologically ordered magnetic  
 1659 phases.

1660 The proposed RAMs formulation successfully extended gradient-based  
 1661 activation mapping to the continuous regression setting through a scale-  
 1662 invariant Gaussian error-decay objective  $\mathcal{O}_p = \exp(-\gamma_p(\theta_p - \hat{\theta}_p)^2)$ , providing  
 1663 physically interpretable spatial attribution maps for each Hamiltonian pa-  
 1664 rameter independently. For the Xception architecture, the layer-resolved  
 1665 RAM analysis revealed a physically consistent hierarchical decoupling across  
 1666 network depth: early entry-flow layers (**block1**, **block3**) capture fine-grained  
 1667 domain periodicity and chiral winding angles at high spatial resolution, while  
 1668 deeper middle-flow and exit-flow layers (**block8**–**block13**) consolidate these  
 1669 local features into coarser, topologically meaningful representations of the  
 1670 global magnetic phase. For the Vision Transformer, the patch-level RAM  
 1671 attributions derived from the reshaped token embedding  $\mathbf{A}_{\text{ViT}} \in \mathbb{R}^{14 \times 14 \times 768}$   
 1672 exhibited smooth, globally coherent spatial patterns that concentrated on  
 1673 chiral domain walls and stripe boundaries in the ordered phases, and pro-  
 1674 duced sparse, peripheral activations for physically degenerate parameter-  
 1675 phase combinations. Critically, both architectures identified the same set of  
 1676 physically meaningful spatial features as the primary drivers of accurate re-  
 1677 gression, confirming that the learned representations are physically grounded  
 1678 rather than statistically arbitrary, and both failed to produce spatially con-  
 1679 centrated activations for the non-identifiable parameters, providing a direct,  
 1680 architecture-agnostic validation of the identifiability hierarchy.

1681 From a deployment standpoint, the inference latency analysis demon-  
 1682 strated that the proposed framework is computationally viable for practical  
 1683 materials characterization workflows. Xception achieves the lowest per-image  
 1684 latency in both the single-sample (127.71 ms) and batch (1.47 ms at  $B = 128$ )  
 1685 regimes, while the Vision Transformer achieves competitive batch throughput

at 1.92 ms/image at  $B = 128$  — a factor of only  $1.31\times$  slower than Xception despite its  $4.1\times$  larger parameter count. These results establish that the Vision Transformer is the preferred architecture for high-throughput batch characterization pipelines, where its superior regression accuracy is achieved at a negligible throughput cost relative to the best CNN baseline.

Taken together, the results of this study demonstrate that regression-based deep learning, combined with physically motivated interpretability methods, constitutes a viable and physically meaningful approach to the inverse Hamiltonian estimation problem in magnetic nanostructures. The proposed RAMs framework is architecture-agnostic and directly applicable to any continuous regression task where spatial attribution of physical parameters is required, extending beyond magnetic nanodots to other physical systems characterized by spatially resolved imaging modalities.

Several directions merit investigation in future work. First, incorporating the full three-dimensional spin vector field  $\mathbf{S}_i = (s_x, s_y, s_z)$  or multi-layer volumetric magnetization images as input would substantially reduce the partial observability constraint and improve the identifiability of the anisotropy and field parameters. Second, domain adaptation strategies, including physics-informed data augmentation and few-shot transfer learning from experimentally labeled samples, are necessary to bridge the simulation-to-experiment gap and enable direct application to real magnetic imaging data. Third, extending the RAM formulation to incorporate uncertainty quantification through ensemble-based or Bayesian attribution methods would provide confidence intervals on the spatial interpretability maps, which are particularly valuable for physically ambiguous parameter-phase combinations near topological phase boundaries. Finally, the integration of the proposed framework with active learning strategies, where the model identifies the most informative parameter combinations to simulate next, could dramatically accelerate the construction of training datasets for high-dimensional Hamiltonian spaces, enabling the extension of the inverse estimation approach to more complex magnetic systems with larger numbers of competing energetic contributions.

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## 1728 Declaration of generative AI and AI-assisted technologies in the 1729 manuscript preparation process

1730 During the preparation of this work the author(s) used Gemini and Claude  
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